

Machine Learning (Mini Project)

Star Type Prediction System

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Introduction

- Stars are described by Temperature, Luminosity, Radius, Spectral Class, and Color.
- Absolute Magnitude (Mv) shows the true brightness of a star.
- Manual calculation is time-consuming and requires expert knowledge.
- Machine Learning helps predict stellar properties faster.
- Our project builds a Star Type Prediction System.
- This is a two-phase project:
 1. Phase 1: Data Cleaning & Regression (Predicting Absolute Magnitude).
 2. Phase 2: Classification & Clustering (Predicting Star Type & Groupings).



Data Cleaning:

Handling missing values, duplicates, outliers



Preprocessing:

Encoding, scaling, feature engineering



Regression:

Predict Absolute Magnitude, Mv



Classification:

Predict Star Type using ML models



Clustering:

Group similar stars

PROBLEM STATEMENT & OBJECTIVES

Problem Statement:

- Manual classification of stars is time-consuming and error-prone.
- Astronomical data is vast and complex, requiring intelligent automation.

Motivation:

- Machine Learning enables efficient analysis of large astronomical datasets.
- Helps identify hidden patterns and improve stellar classification accuracy.

Objectives:

1. Analyze relationships among stellar parameters.
2. Predict continuous attributes using regression.
3. Classify stars into their respective types.
4. Perform clustering for pattern recognition.
5. Compare model performance using metrics.
6. Visualize and interpret key findings.

Dataset Overview

- **Source:** star_type_dataset.csv
- **Shape:** 15,000 rows × 14 columns
- **Features include:**
 1. **Numerical:** Temperature (K), Luminosity (L/Lo), Radius (R/Ro), Absolute Magnitude (Mv), Age, Metallicity, Rotational Velocity, Star Type ID, Distance
 2. **Categorical:** Color, Spectral Class, Galaxy, Constellation, Star Type
- **Target Variable:**
 - Absolute Magnitude (Mv) (for regression)
 - Star Type (for classification)
- Some columns have missing values (Color, Spectral Class, Age, Metallicity, Rotational Velocity, Distance).
- Wide range of values observed (e.g., Temperature: 800K – 49,776K, Luminosity varies up to 10^9 L/Lo).

Dataset Overview

Sample Dataset Records (First 5 Rows)

	Temperature (K)	Luminosity (L/L _o)	Radius (R/R _o)	Absolute Magnitude (M _v)	Color	Spectral Class	Galaxy	Constellation	Age (in million years)	Metallicity ([Fe/H])	Rotational Velocity (km/s)	Distance (light years)	Star Type ID	Star Type
0	3882	0.002749	0.116150	11.232131	Orange	K	Andromeda	Cygnus	2072.429337	-0.831932	9.914174	6279.918705	1	Red Dwarf
1	4866	0.234267	0.682442	6.405724	Orange	K	Milky Way	Ursa Major	8061.732185	0.399705	8.937917	399.135195	3	Main Sequence
2	8340	0.001363	0.017720	11.993847	White	A	Triangulum	Cassiopeia	611.833850	-0.436312	205.773440	2056.084304	2	White Dwarf
3	7176	3.828776	1.268587	3.372350	Yellow-White	F	Milky Way	Pegasus	1549.733369	-0.472496	90.250004	4889.970842	3	Main Sequence
4	6412	37938.044899	158.163228	-6.617687	Yellow-White	F	Milky Way	Lyra	185.828099	-0.220985	67.493993	57183.271180	4	Supergiant

Summary of Key Numerical Features

	Temperature (K)	Luminosity (L/L _o)	Radius (R/R _o)	Absolute Magnitude (M _v)	Age (in million years)
count	15000.000000	1.500000e+04	15000.000000	15000.000000	14700.000000
mean	5738.729600	2.824617e+06	62.489976	6.216801	5616.860567
std	4654.715091	3.791049e+07	209.684482	7.177768	4143.699246
min	800.000000	1.980002e-06	0.008004	-18.103157	1.012774
25%	3214.000000	5.192790e-03	0.231031	4.442248	1704.508004
50%	4552.000000	3.148912e-02	0.488588	8.584599	5386.787196
75%	6623.000000	1.429226e+00	1.174739	10.541498	9284.011392
max	49776.000000	1.490263e+09	1997.561669	19.088336	12999.529827

Data Preprocessing & Cleaning

- **Missing Values Handling:**

- 1. Missing Value Treatment:

- **Technique**

- Technique 1: Deletion → Removed 300 rows → Data loss → Not used
 - Technique 2: KNN Imputation → Accurate but complex
 - Technique 3: Median/Mode Imputation → Selected Method (No data loss)

- 2. **Result:** No missing values remaining.

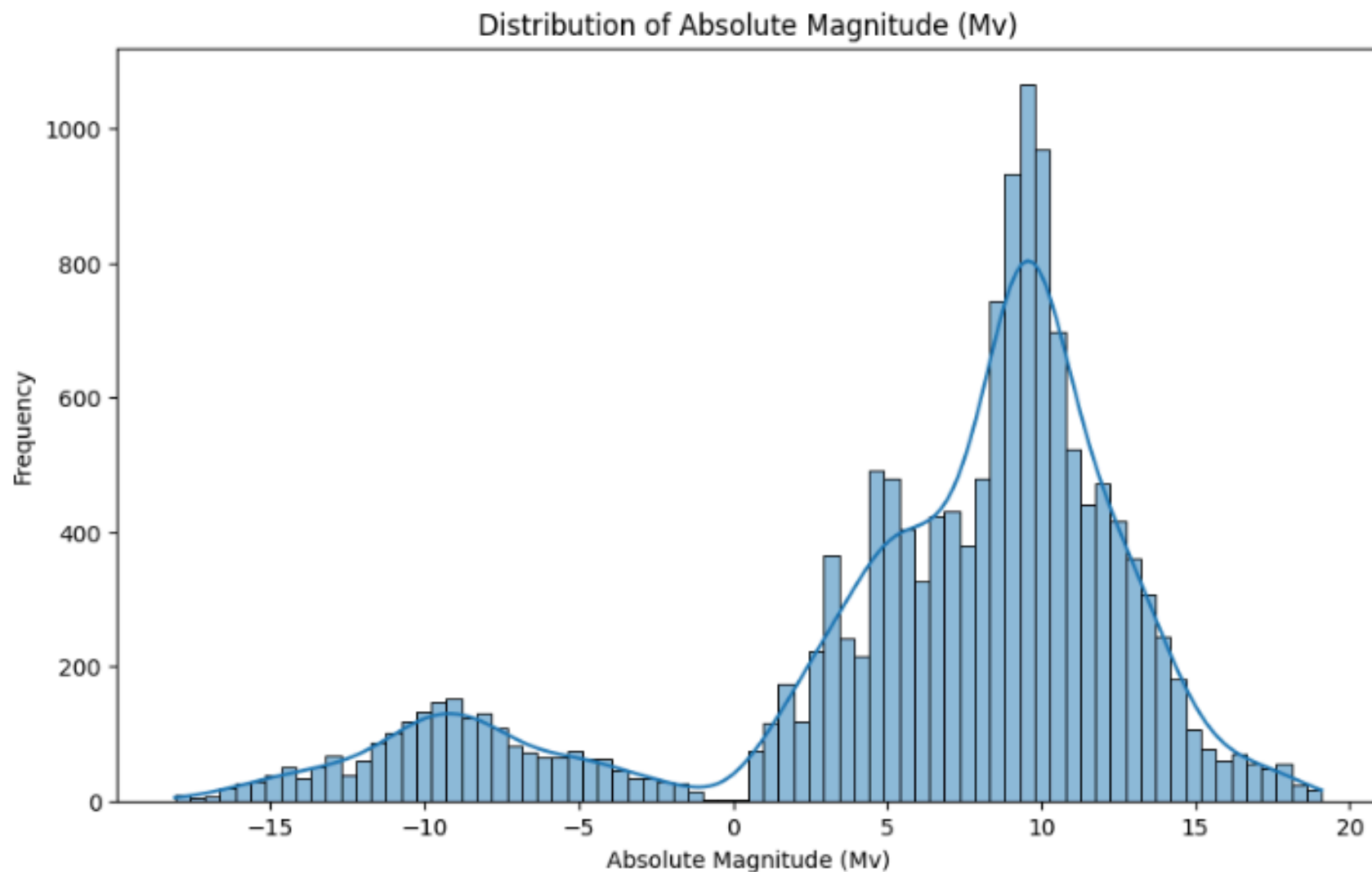
- **Duplicate Rows:**

Checked for duplicates → 0 duplicates found.

- **Outcome:** Clean dataset ready for analysis and modeling.

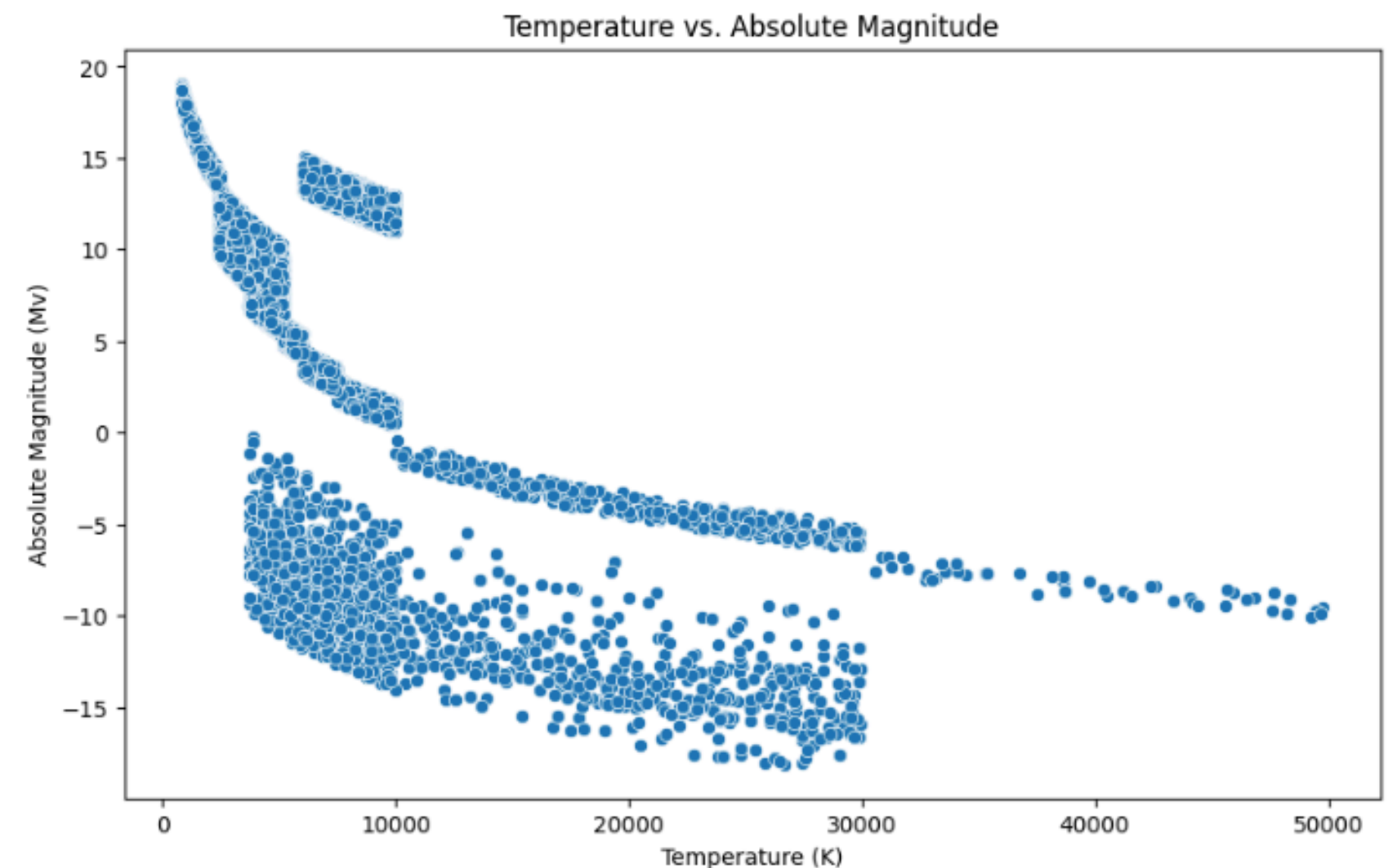
Technique Used	Data Loss	Accuracy of Estimation	Complexity	Suitability
Deletion	High	Low	Low	Simple datasets with few missing
KNN Imputation	None	High	High	Datasets with numeric similarity relations
Median/Mode Imputation	None	Moderate to High	Low	Balanced approach suitable for mixed data

Feature Distributions & Relationships

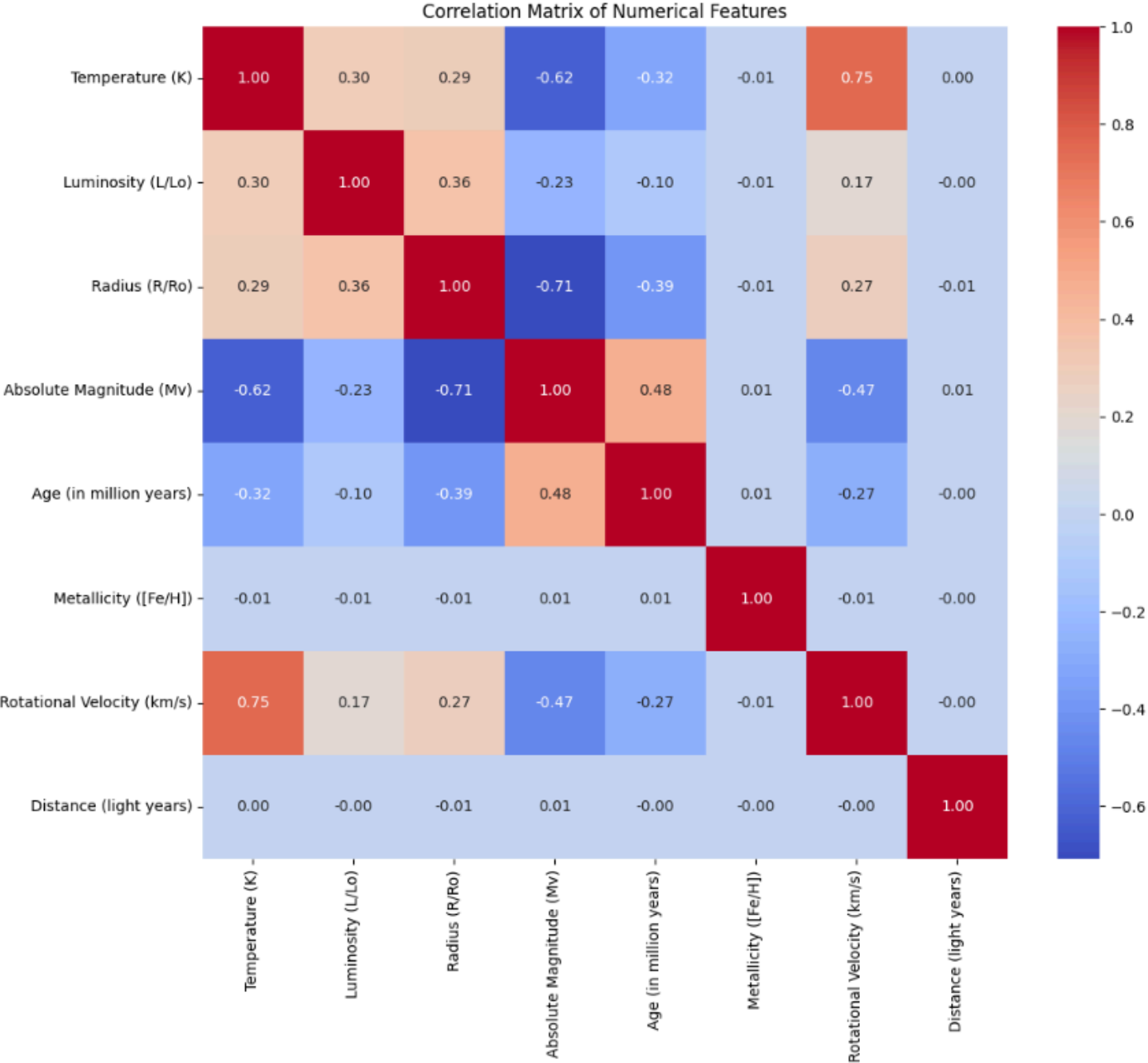


TARGET VARIABLE DISTRIBUTION:
HISTOGRAM OF ABSOLUTE MAGNITUDE (MV) WITH KDE
SHOWS THE SPREAD OF VALUES.

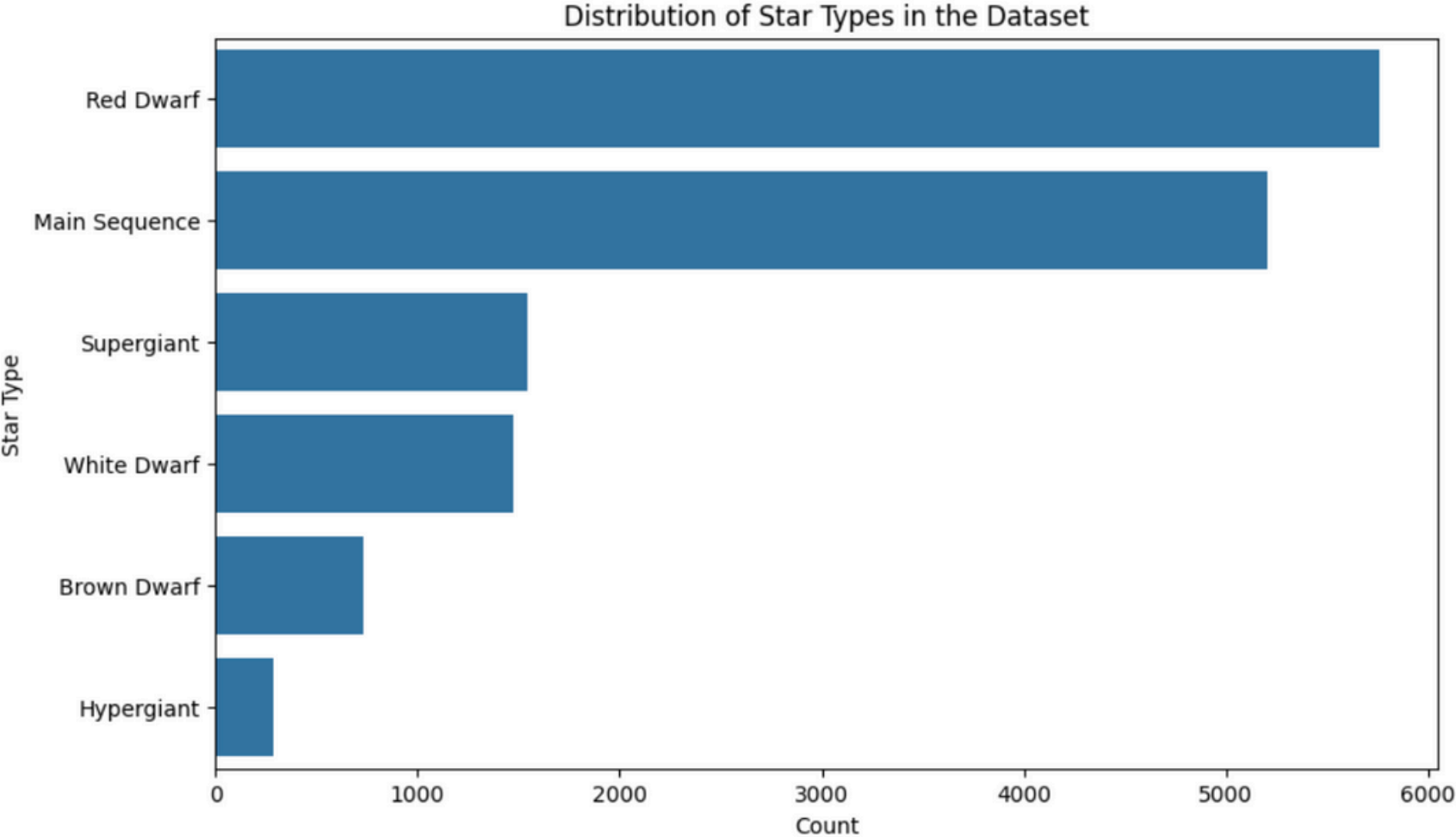
FEATURE RELATIONSHIPS:
SCATTER PLOT: TEMPERATURE VS ABSOLUTE MAGNITUDE
SHOWS HIGHER TEMPERATURE → LOWER MAGNITUDE
TREND.



Feature Distributions & Relationships



STAR TYPE DISTRIBUTION:
COUNTPLOT SHOWS THE FREQUENCY OF DIFFERENT STAR
TYPES IN THE DATASET.



- CORRELATION ANALYSIS:
- CORRELATION HEATMAP OF NUMERICAL FEATURES.
 - EXAMPLE: AGE IS STRONGLY CORRELATED WITH ABSOLUTE MAGNITUDE (MV).

Outlier Detection and Removal

Method	Rows Retained	Mean	Median	Std Dev
Original	15000	2.13745	0.031004	4.576472
IQR	12294	0.221414	0.015854	0.453107
Z-Score	14639	1.755293	0.028111	3.918343
Isolation Forest	14250	1.583229	0.025267	3.767415

- **Goal:** Detect and remove abnormal values to improve data reliability.
- **Methods Applied:**
 1. Log transformation of Luminosity (L/Lo) → $\text{Luminosity_log} = \log_{10}(\text{Luminosity})$
 2. **IQR Method:** retained 12,294 of 15,000 rows (-18%)
 3. **Z-Score Method:** retained 14,639 rows ($|Z| > 3$)
 4. **Isolation Forest:** retained 14,250 rows — balanced filtering
- **Result:** Isolation Forest chosen → best balance between data retention and outlier removal.

Feature Encoding, Train–Test Split & Scaling

- **Feature Encoding:**

- Converted categorical features (Color, Spectral Class, Galaxy, Constellation) using One-Hot Encoding
- Expanded into 38 numerical attributes
- Target Variable: Absolute Magnitude (Mv)

- **Train–Test Split (80–20):**

Dataset	Rows	Columns
X_train	11,400	38
X_test	2,850	38
y_train	11,400	–
y_test	2,850	–

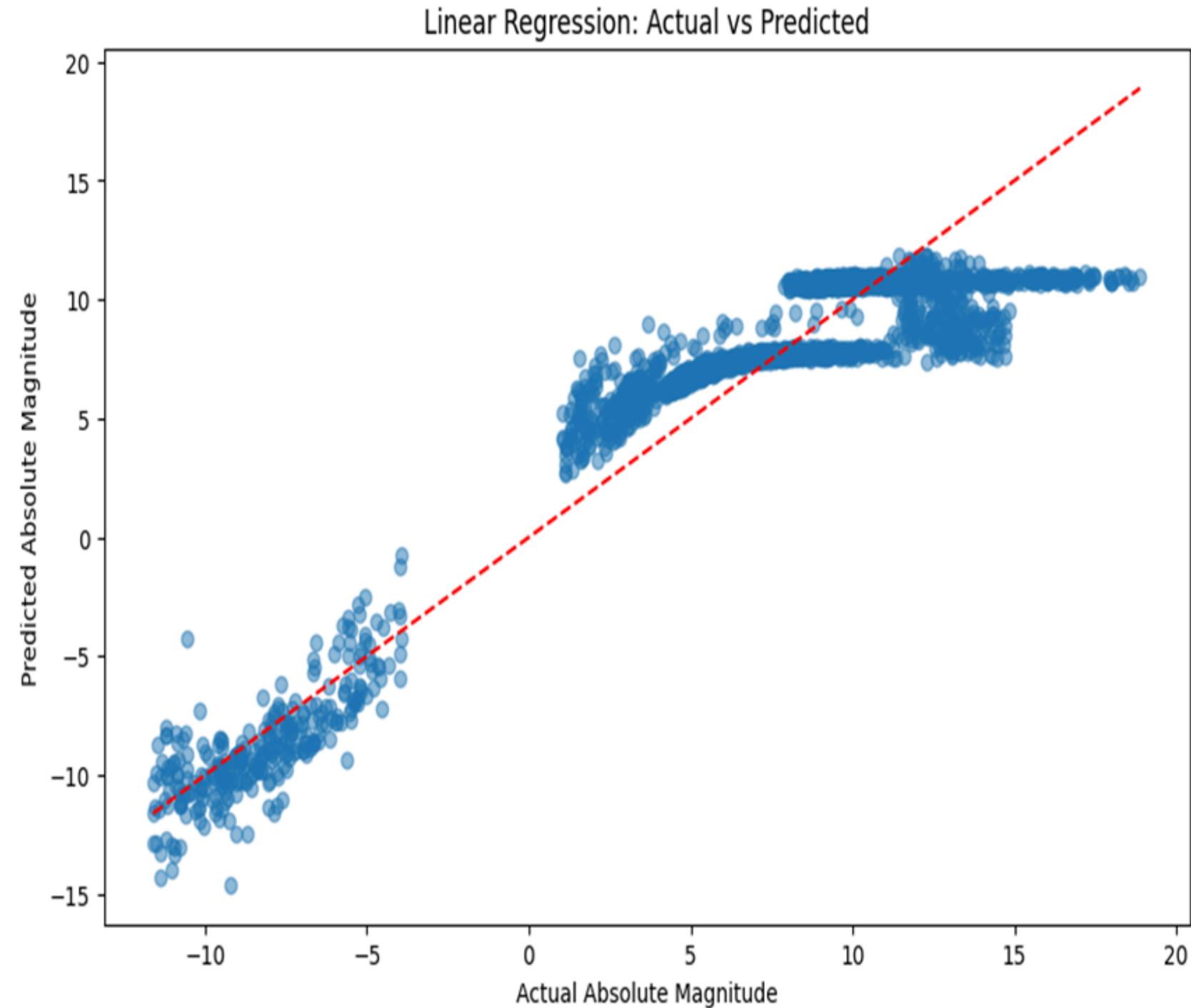
- **Feature Scaling:**

- **StandardScaler:** Mean = 0, Std = 1 (normal distribution)
- **MinMaxScaler:** Scales 0–1 (preserves feature shape)
- **RobustScaler:** Uses median & IQR (best for outliers)

- **Observation:**

- Both train/test sets preserve data distribution
- All scaled datasets: (11,400 × 38)
- RobustScaler produced most consistent results after outlier removal

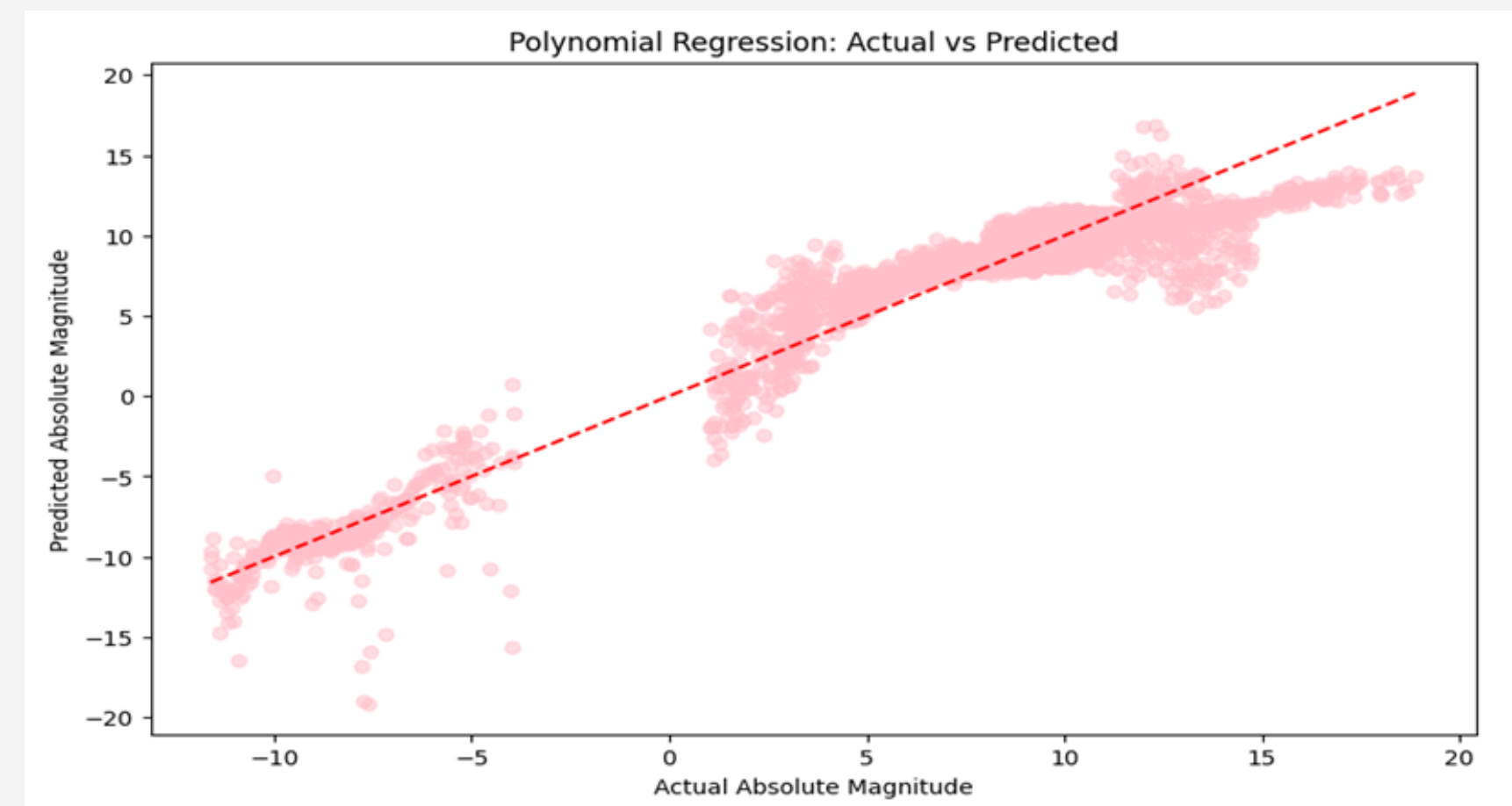
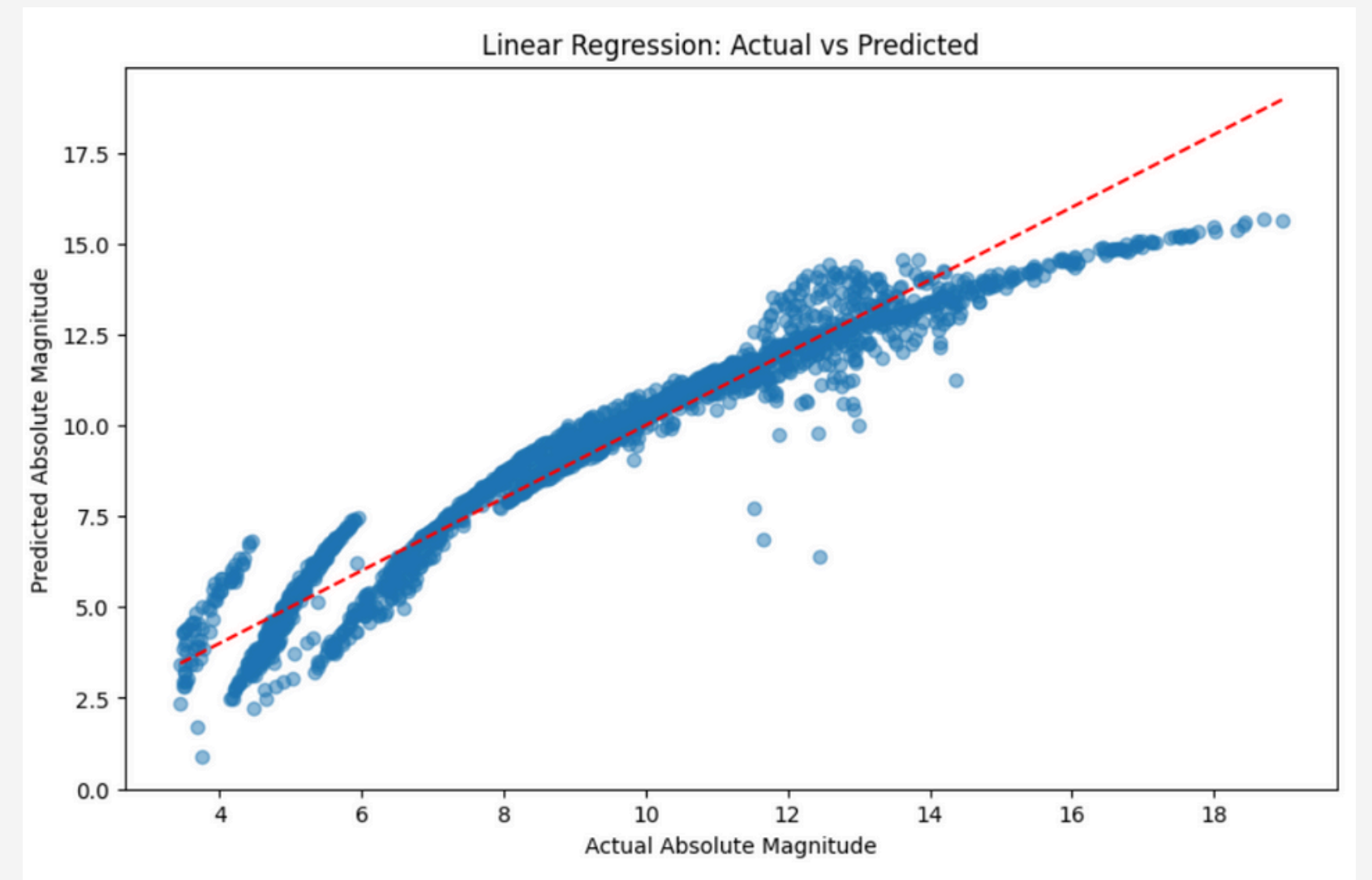
Regression Models: Implementation & Evaluation



- Regression models were used to predict Absolute Magnitude (M_v) of stars.
- **Evaluation metrics:** MAE, MSE, RMSE, R^2 .
- **Goal:** Identify the best-performing algorithm for stellar magnitude prediction.
- **Models implemented:** Linear, Polynomial, Ridge, Lasso, ElasticNet, RANSAC, Decision Tree, Random Forest.

Linear & Polynomial Regression

- **Linear Regression:**
 - $R^2 = 0.87$, explained 87% of variance.
 - Minor deviations observed — could not capture all non-linear trends.
- **Polynomial Regression (Degree = 2):**
 - Improved $R^2 = 0.90$, MAE = 1.41.
 - Captured non-linear patterns more effectively.



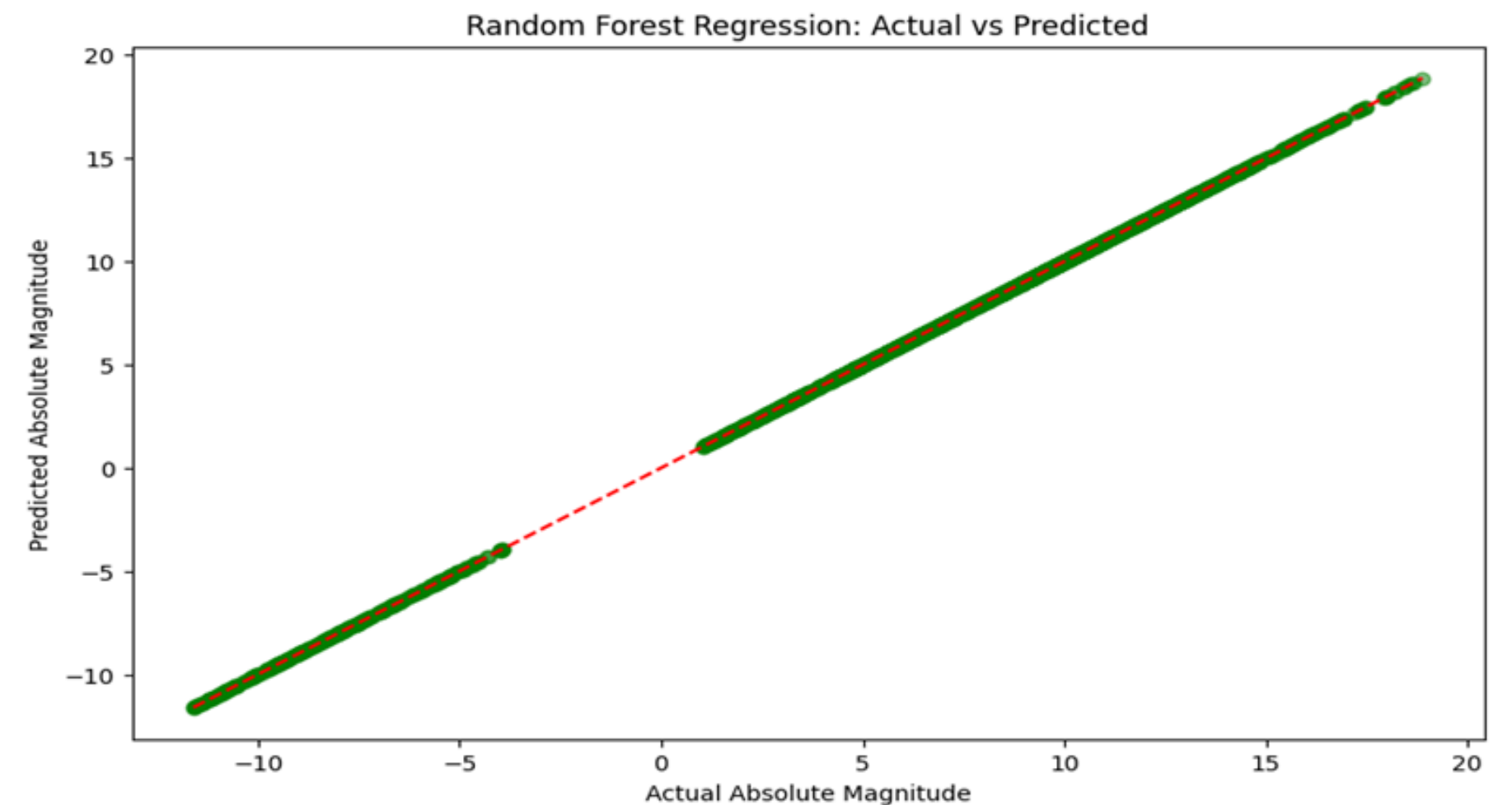
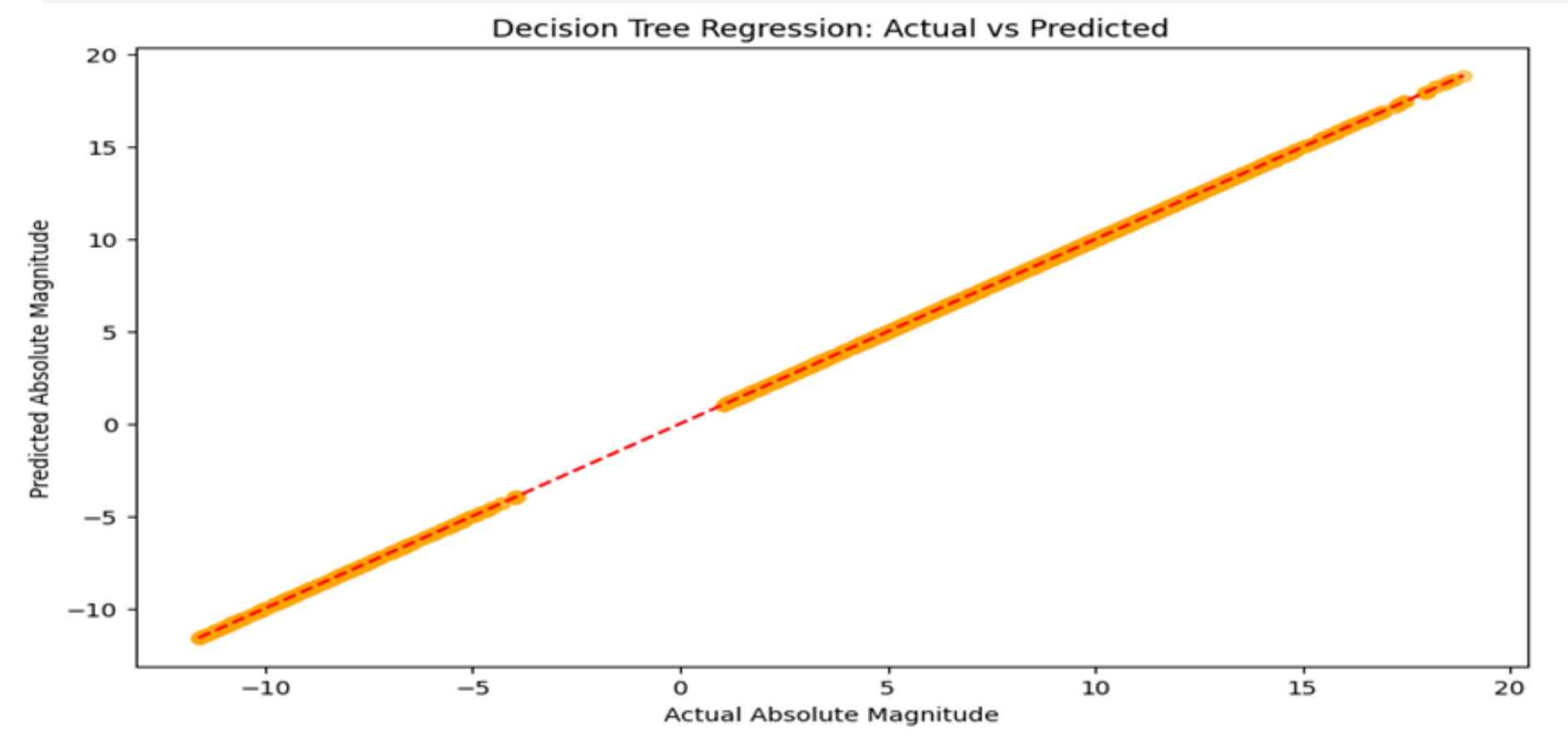
Regularized & Robust Models

Model	R ² Score	MAE	MSE	RMSE	Remarks
Linear Regression	0.8721	1.73	4.93	2.22	Baseline model
Ridge Regression	0.8721	1.73	4.93	2.22	No improvement over Linear
Lasso Regression	0.8719	1.73	4.94	2.22	Similar to Ridge
ElasticNet Regression	0.8719	1.73	4.94	2.22	Similar results
RANSAC Regression	-27199	Very High	Very High	Very High	Failed due to no strong outliers

- **Ridge, Lasso, ElasticNet Regression:**
 - $R^2 \approx 0.87$ each.
 - Regularization did not improve accuracy → low multicollinearity.
- **RANSAC Regression:**
 - $R^2 = -27199$, very high error values.
 - Failed due to absence of strong outliers.

Decision Tree & Random Forest

- **Decision Tree Regression:**
 - $R^2 = 0.999999$, $MAE \approx 0.0019$.
 - Excellent fit, but possible overfitting risk.
- **Random Forest Regression:**
 - $R^2 = 1.0000$, $MAE = 0.0012$, $RMSE = 0.0025$.
 - Best accuracy and generalization via ensemble averaging.
 - Selected as final model for prediction.

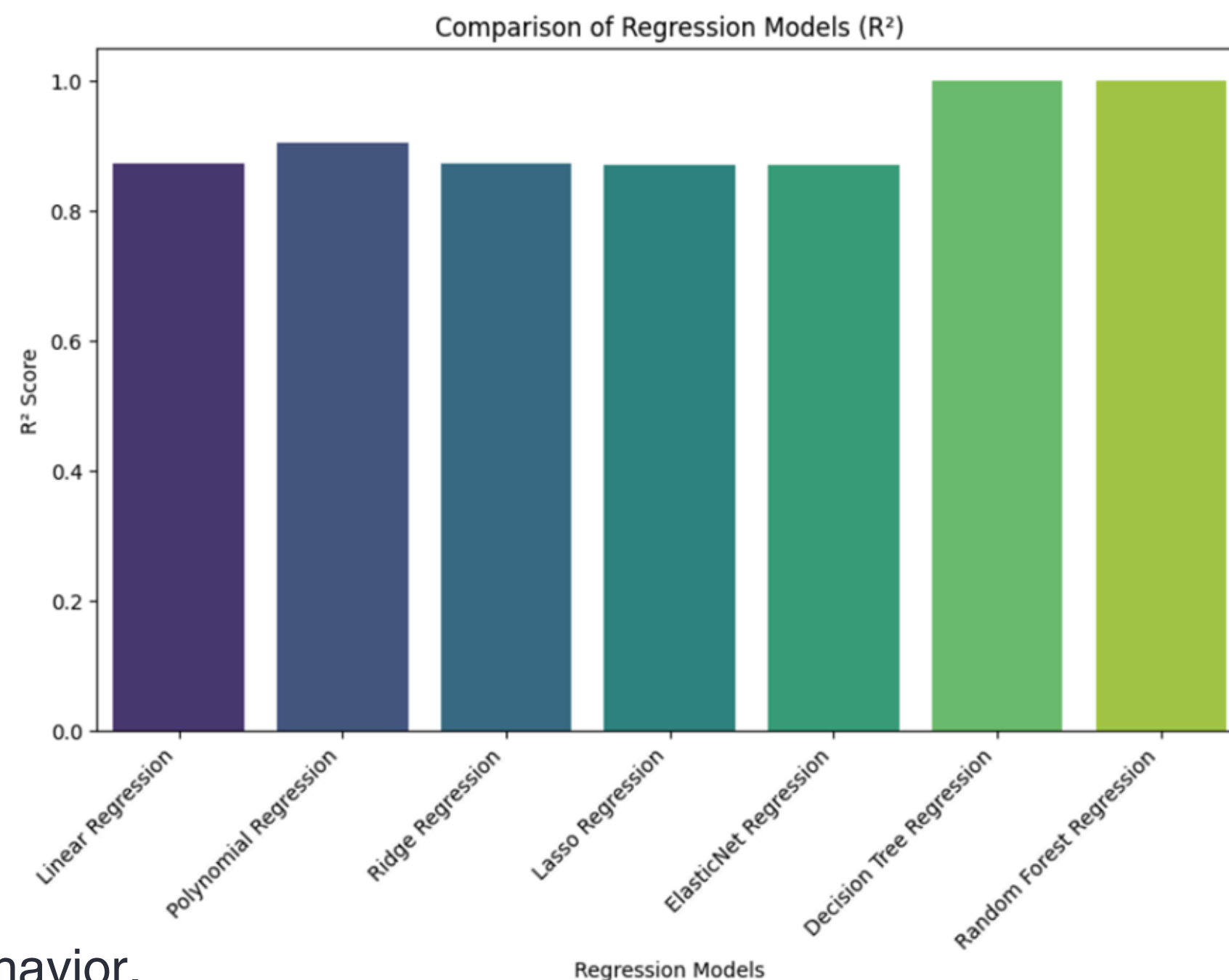


Model Comparison & Conclusion

Model	R ² Score	MSE	MAE	RMSE
Linear Regression	0.8721	4.93	1.73	2.22
Polynomial Regression	0.9055	3.64	1.41	1.9
Ridge Regression	0.8721	4.93	1.73	2.22
Lasso Regression	0.8719	4.94	1.73	2.22
ElasticNet Regression	0.8719	4.94	1.73	2.22
Decision Tree Regression	0.99999	0.00002	0.0019	0.0046
Random Forest	1	0.000006	0.0012	0.0025

Key Takeaways:

- Dataset shows strong non-linear behavior.
- Polynomial & Tree-based models outperformed linear ones.
- Random Forest Regression gave the best and most stable results.
- Preprocessing, encoding, and scaling contributed to optimal model performance.



Classification Models: Overview & Data Preparation

- **Objective:**

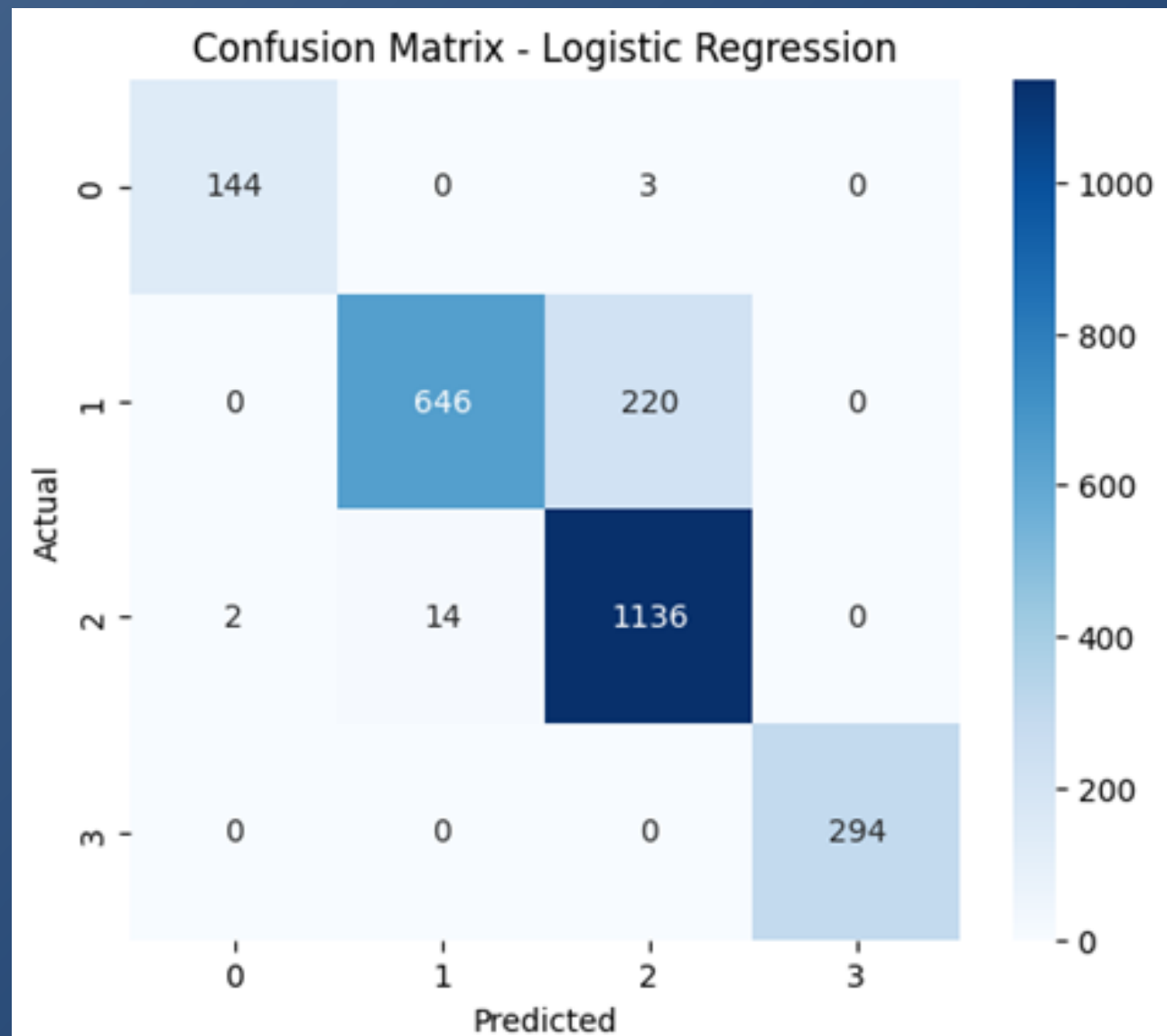
- Predict Star Type using various machine learning classification algorithms.

- **Key Features Used:**

- Temperature
- Luminosity
- Radius
- Color
- Spectral Class

- **Data Preparation:**

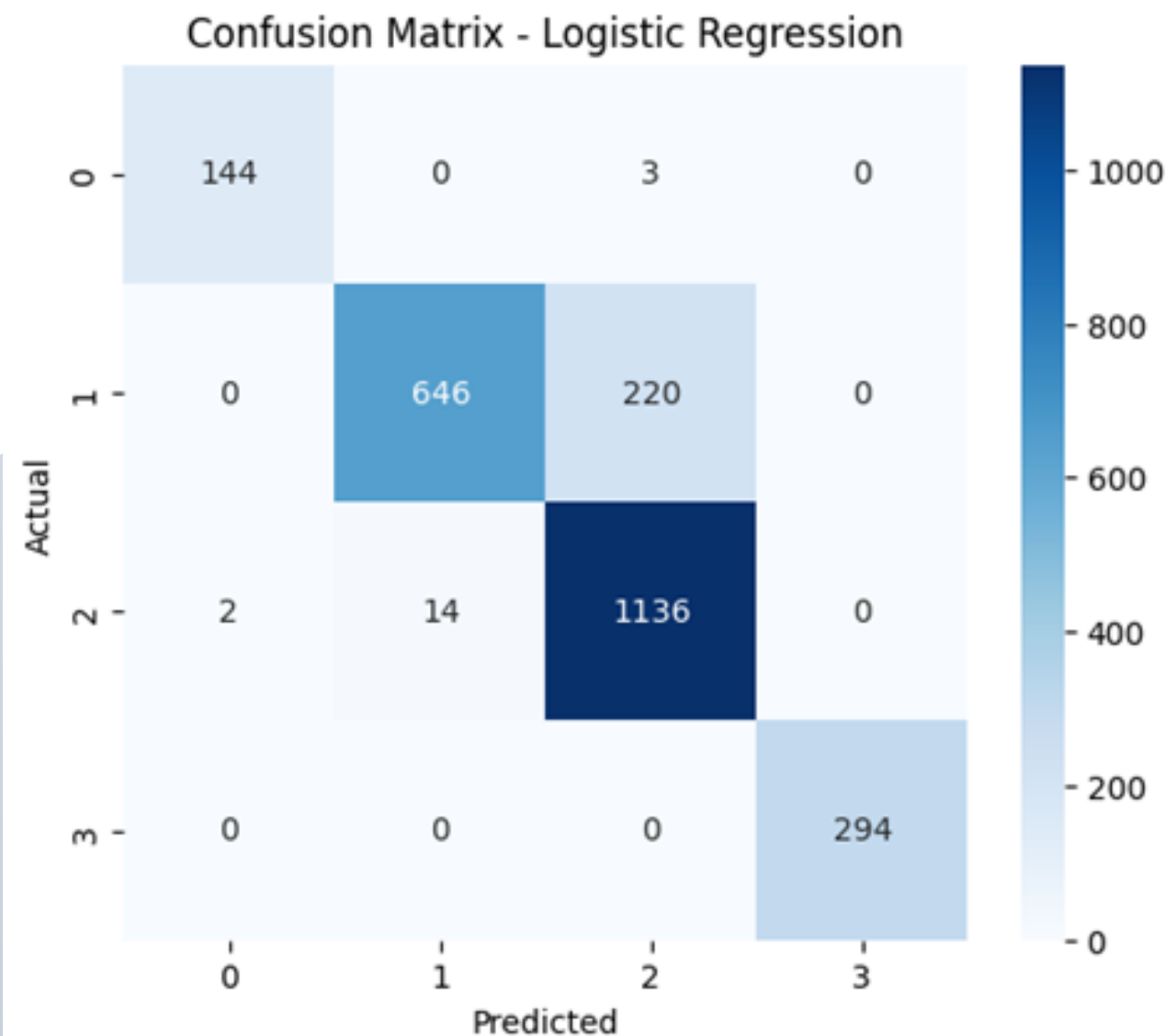
- One-Hot Encoding: Converted categorical columns (Color, Spectral Class, Galaxy, Constellation) → 32 features.
- Train-Test Split: 80 : 20 with Stratified Sampling.
- Feature Scaling: StandardScaler for normalization.
- Ensured balanced and clean dataset for fair model training.



Logistic Regression & Decision Tree

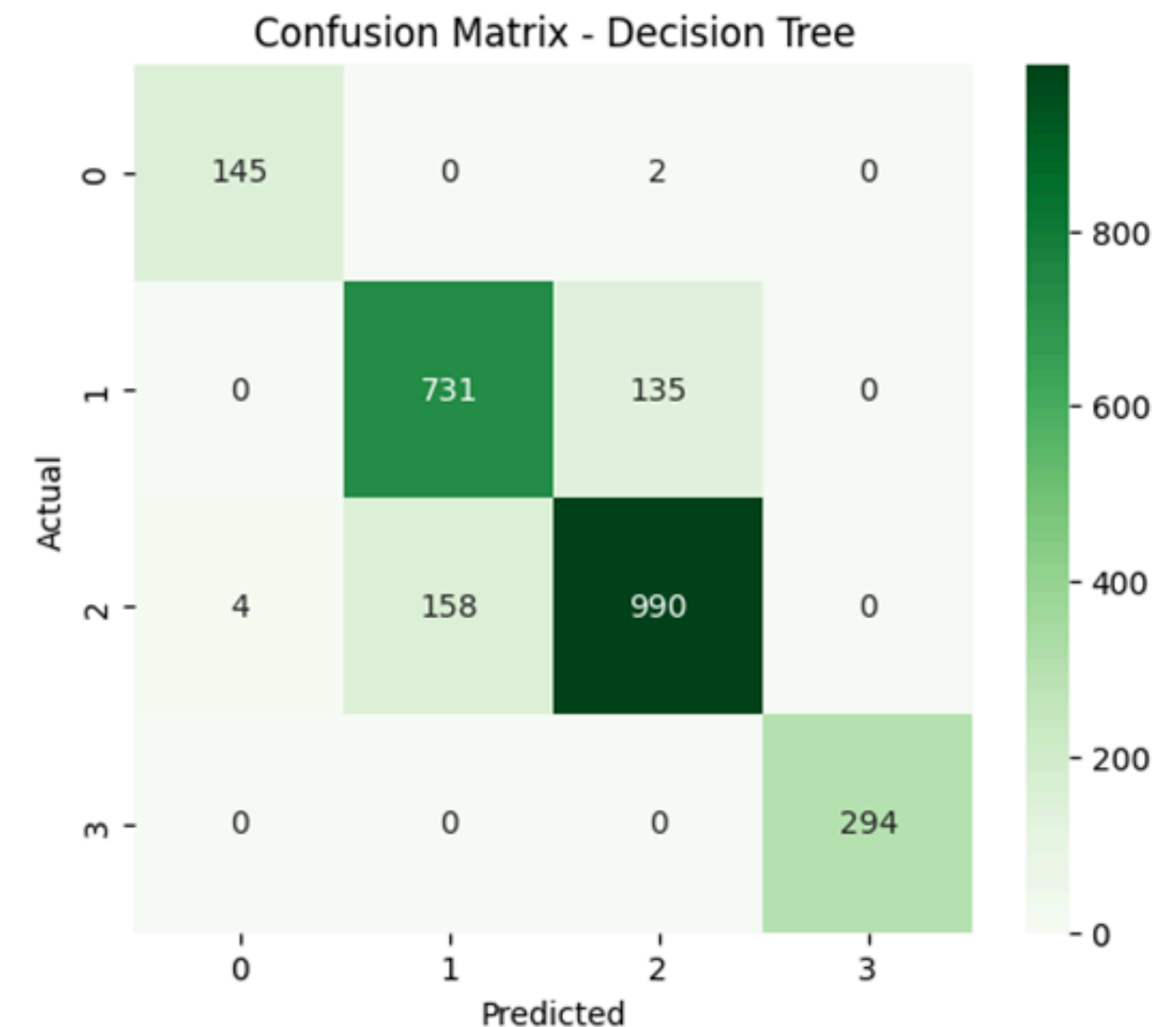
Logistic Regression:

- Linear decision boundary model.
- Accuracy = 90.28 %.
- Precision $\approx$ 0.91 | Recall $\approx$ 0.90 | F1 $\approx$ 0.90.
- Balanced classification; minimal misclassifications.

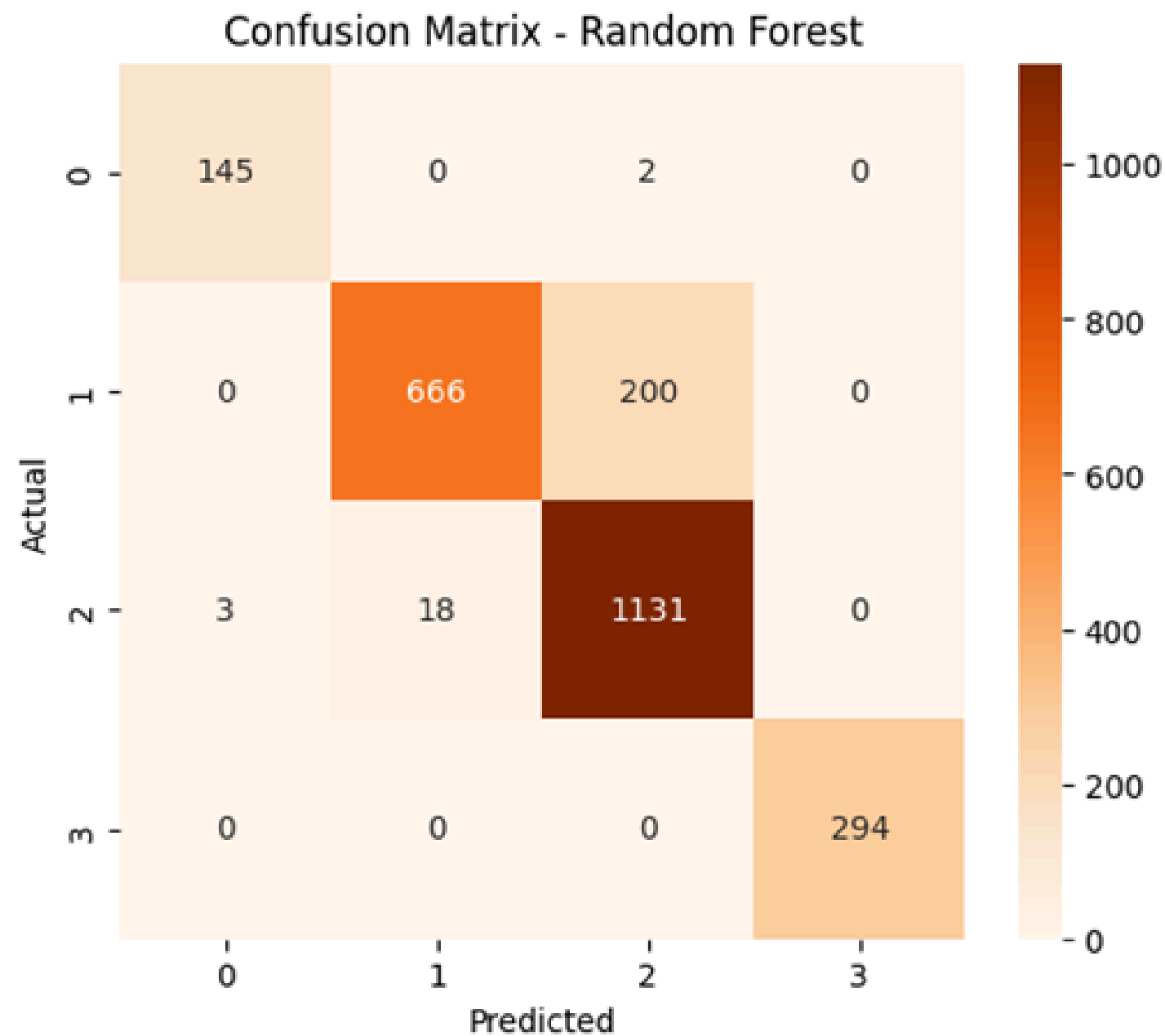


Decision Tree Classifier:

- Captured non-linear relationships.
- Accuracy = 87.84 %.
- Slight overfitting; interpretability high.
- Class overlap caused minor misclassifications.



Random Forest Classifier

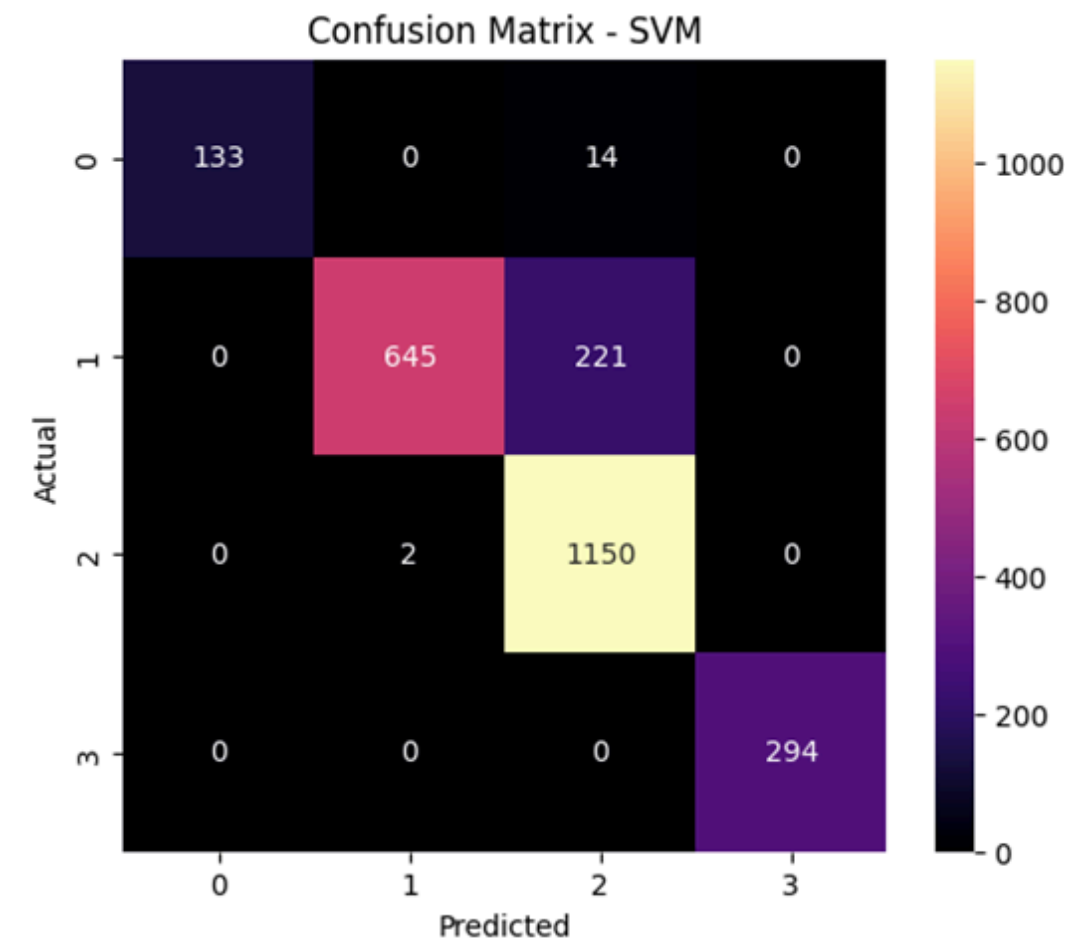
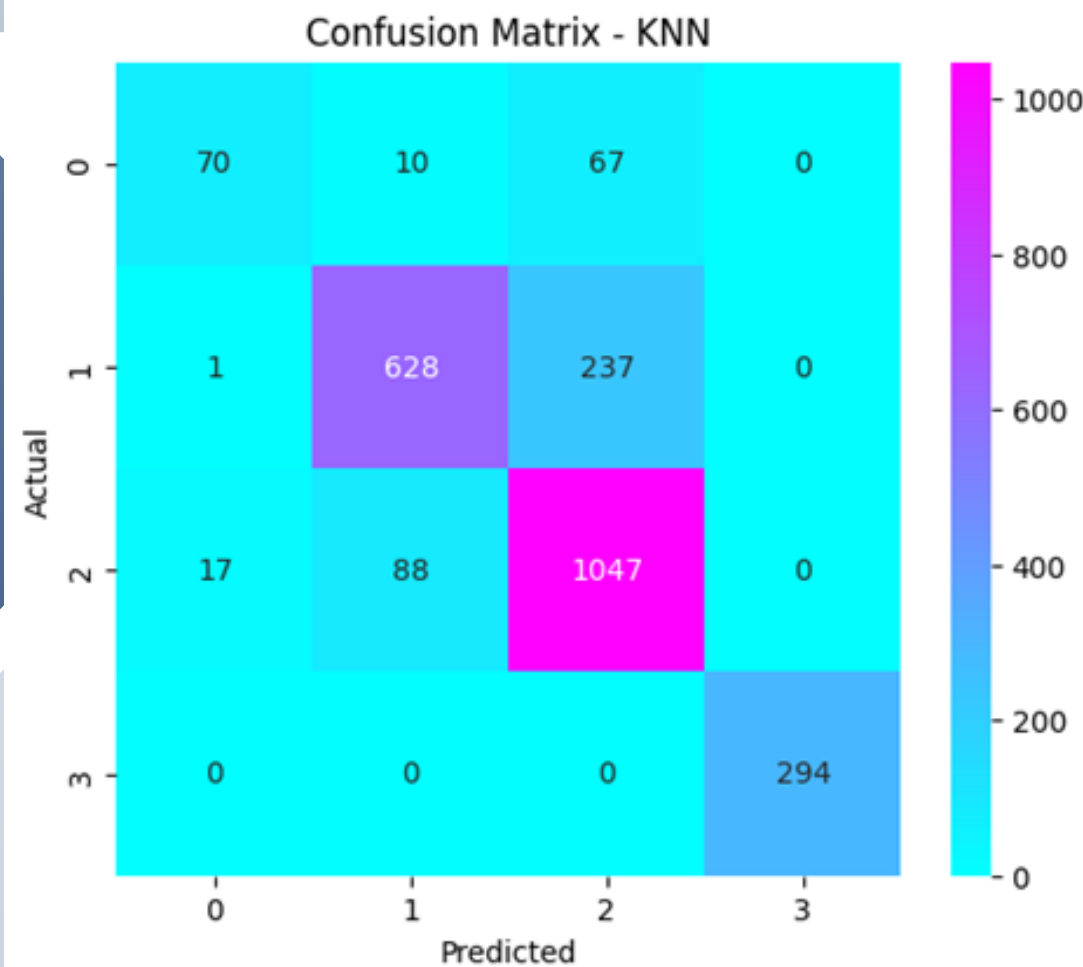


- **Model Description:**
 - Ensemble of multiple Decision Trees to reduce overfitting.
- **Results:**
 - Accuracy = 90.93 % (highest among all).
 - Precision = 0.9185 | Recall = 0.9093 | F1 = 0.9074.
 - Robust, consistent performance across classes.
 - Provided feature importance insights.

KNN, SVM, Naive Bayes

K-Nearest Neighbors (KNN):

- Accuracy = 82.91 % (lowest).
- Sensitive to high dimensions & distance metrics.
- Struggled with overlapping classes.

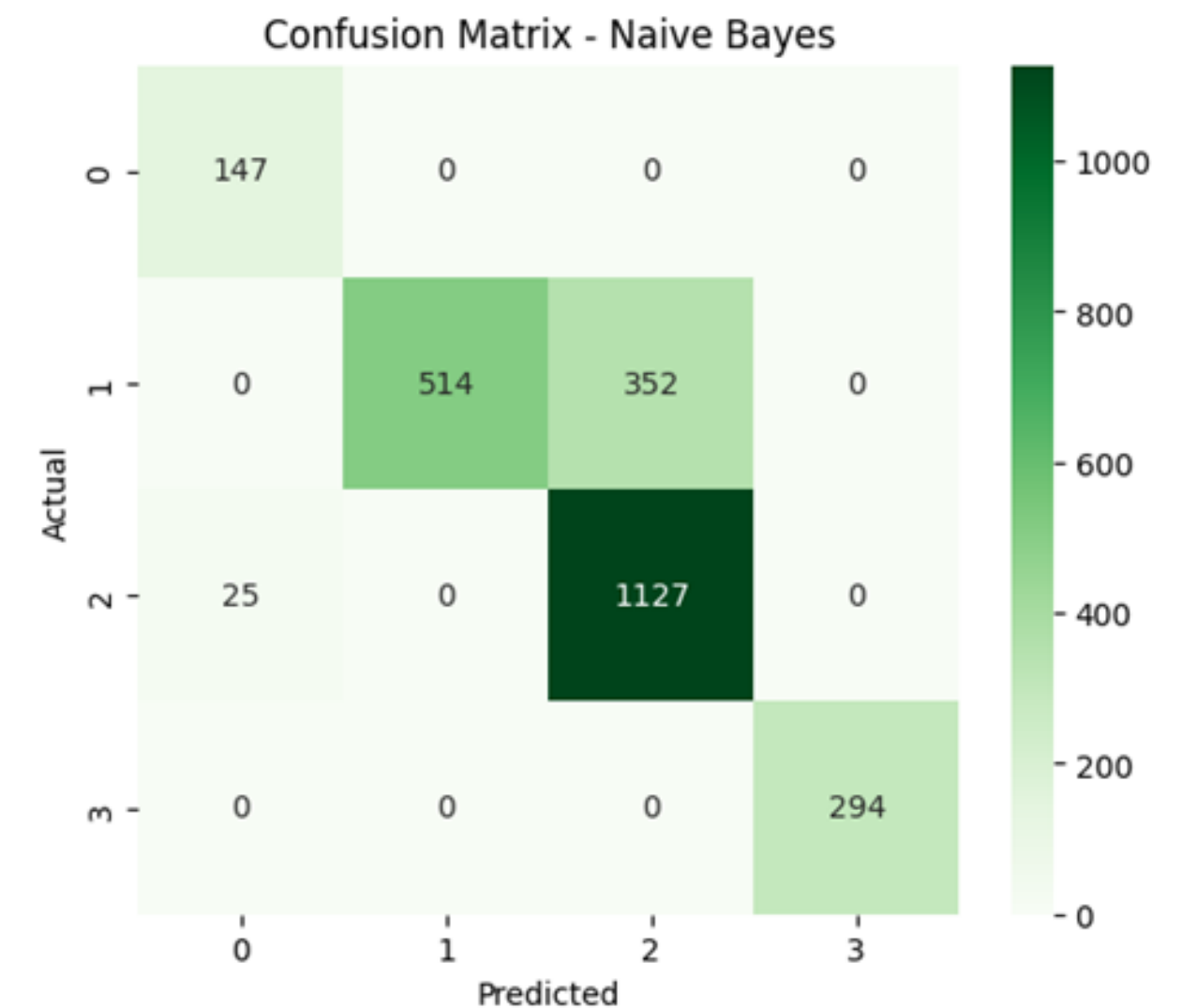


Support Vector Machine (SVM):

- Accuracy = 90.36 %.
- Kernel-based model handled non-linear boundaries.
- High generalization and precision.

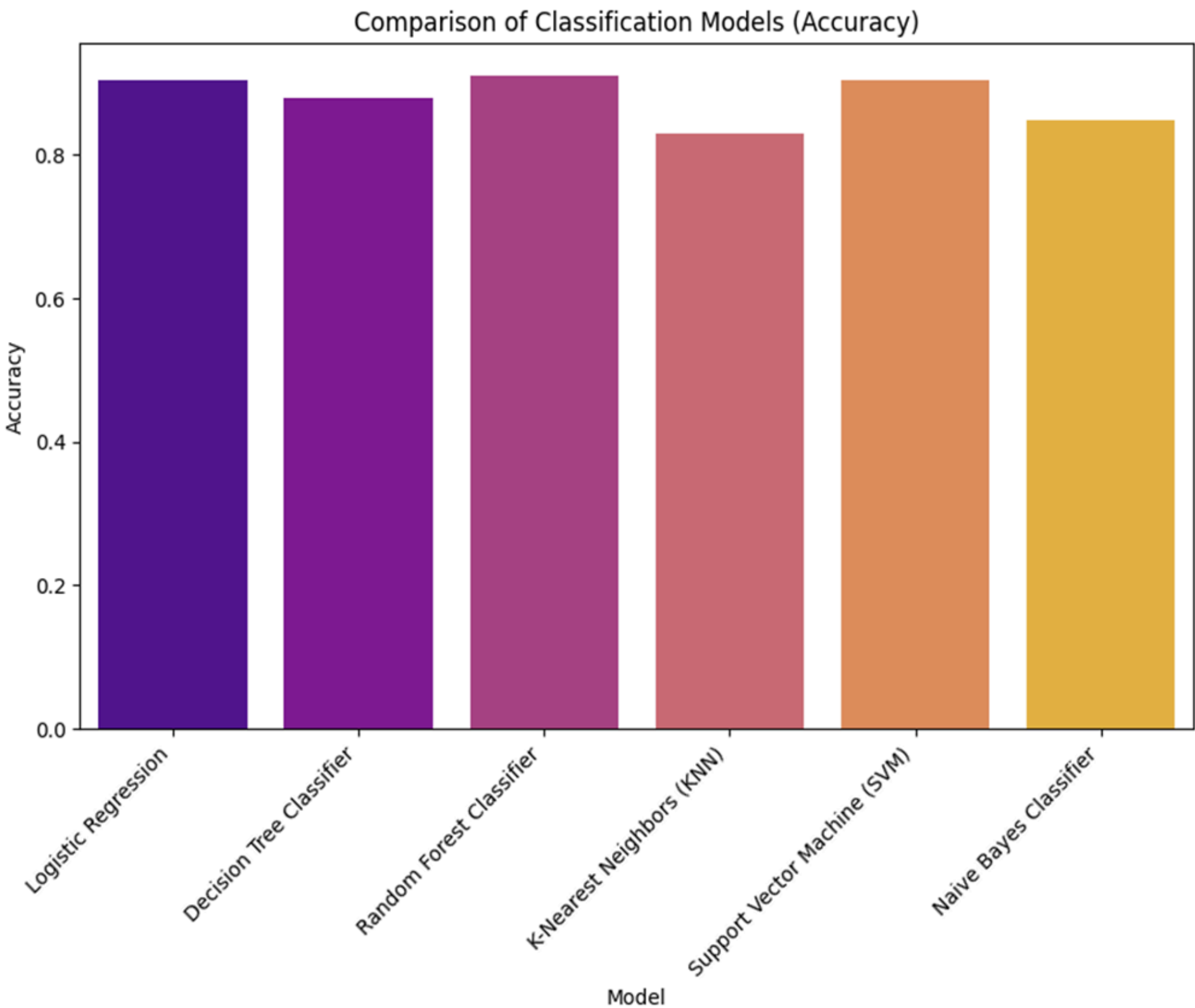
Naive Bayes Classifier:

- Accuracy = 84.66 %.
- Fast and simple baseline model.
- Performed poorly with correlated features.



Comparative Analysis of Models

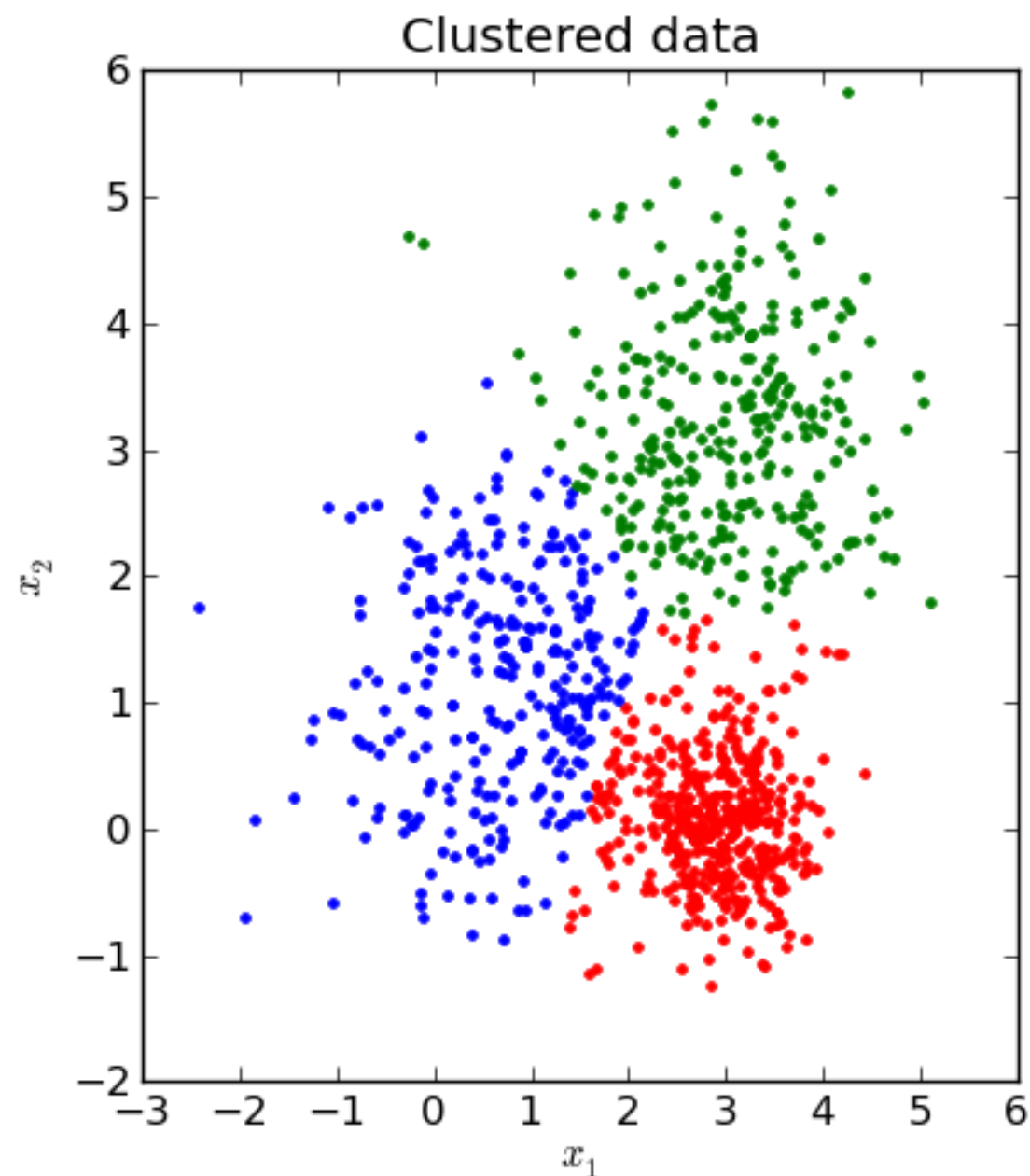
Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9028	0.9148	0.9028	0.9003
Decision Tree Classifier	0.8784	0.8788	0.8784	0.8785
Random Forest Classifier	0.9093	0.9185	0.9093	0.9074
KNN Classifier	0.8291	0.8348	0.8291	0.8249
SVM Classifier	0.9036	0.9194	0.9036	0.9013
Naive Bayes Classifier	0.8466	0.8798	0.8466	0.8383



Observations

- Random Forest delivered the best overall accuracy (90.93 %) and strongest generalization.
- SVM & Logistic Regression showed stable, balanced performance.
- Decision Tree & Naive Bayes achieved moderate accuracy.
- KNN was least effective in high-dimensional space.
- Ensemble and kernel-based models clearly outperformed others.
- Proper encoding, scaling, and preprocessing ensured optimal results.
- All models achieved satisfactory accuracy, confirming the dataset's quality.

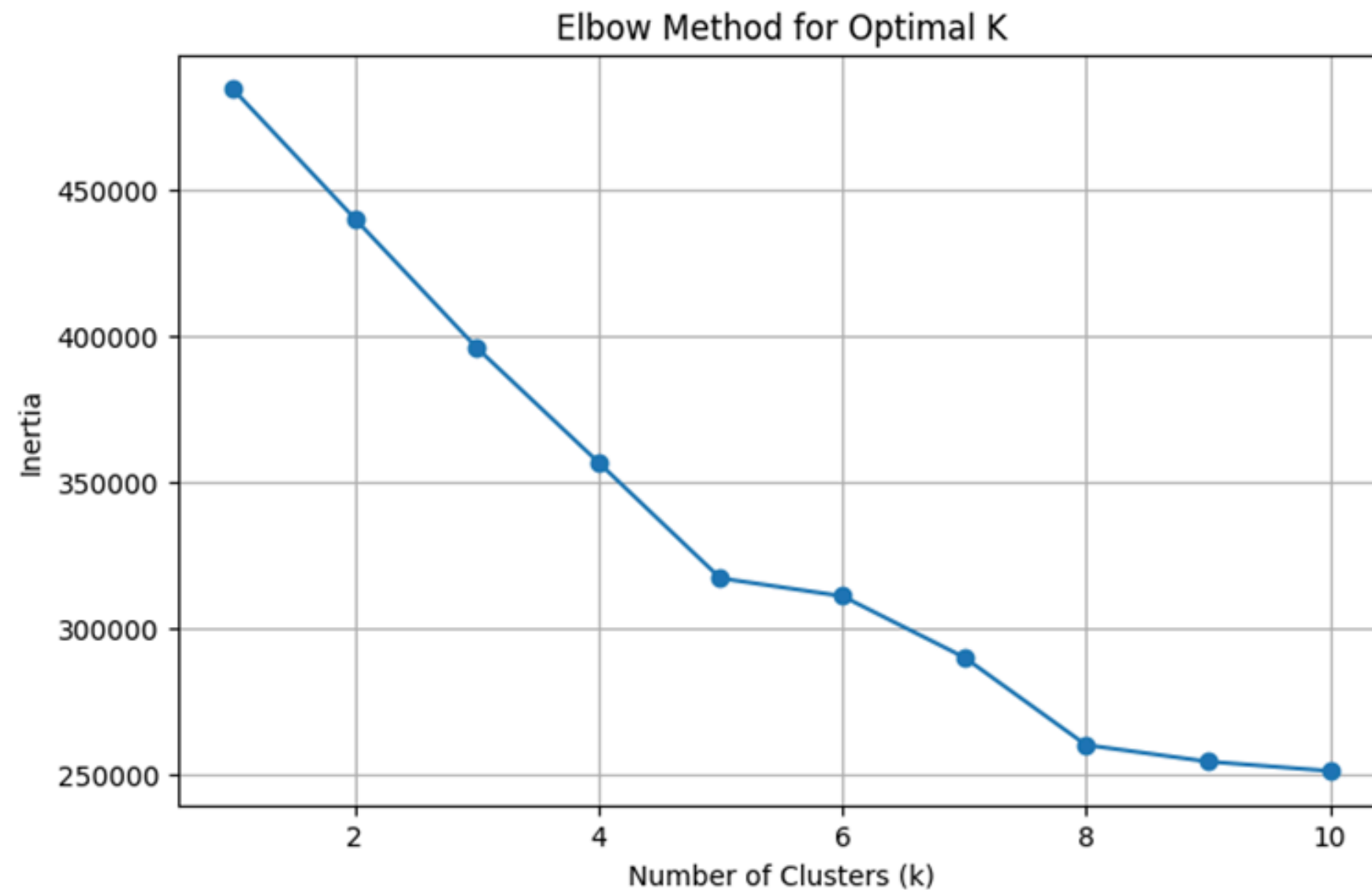
Clustering Models: Introduction & Data Preparation



- **Objective:**
 - To group stars based on their intrinsic features (Temperature, Luminosity, Radius, Spectral Type) and identify natural clusters.
- **Data Preparation:**
 - Removed irrelevant columns: Star Type ID, Star Type
 - Applied One-Hot Encoding → 35 total features
 - Performed Standard Scaling to normalize data
 - Ensured all features contributed equally to clustering
- **Goal:**
 - Detect distinct star groupings through unsupervised learning.

Determining Optimal Clusters (K Value)

- Elbow Method:
 - Plotted Inertia vs K
 - Clear elbow at K = 5 → optimal cluster number
 - Indicates minimal gain beyond 5 clusters

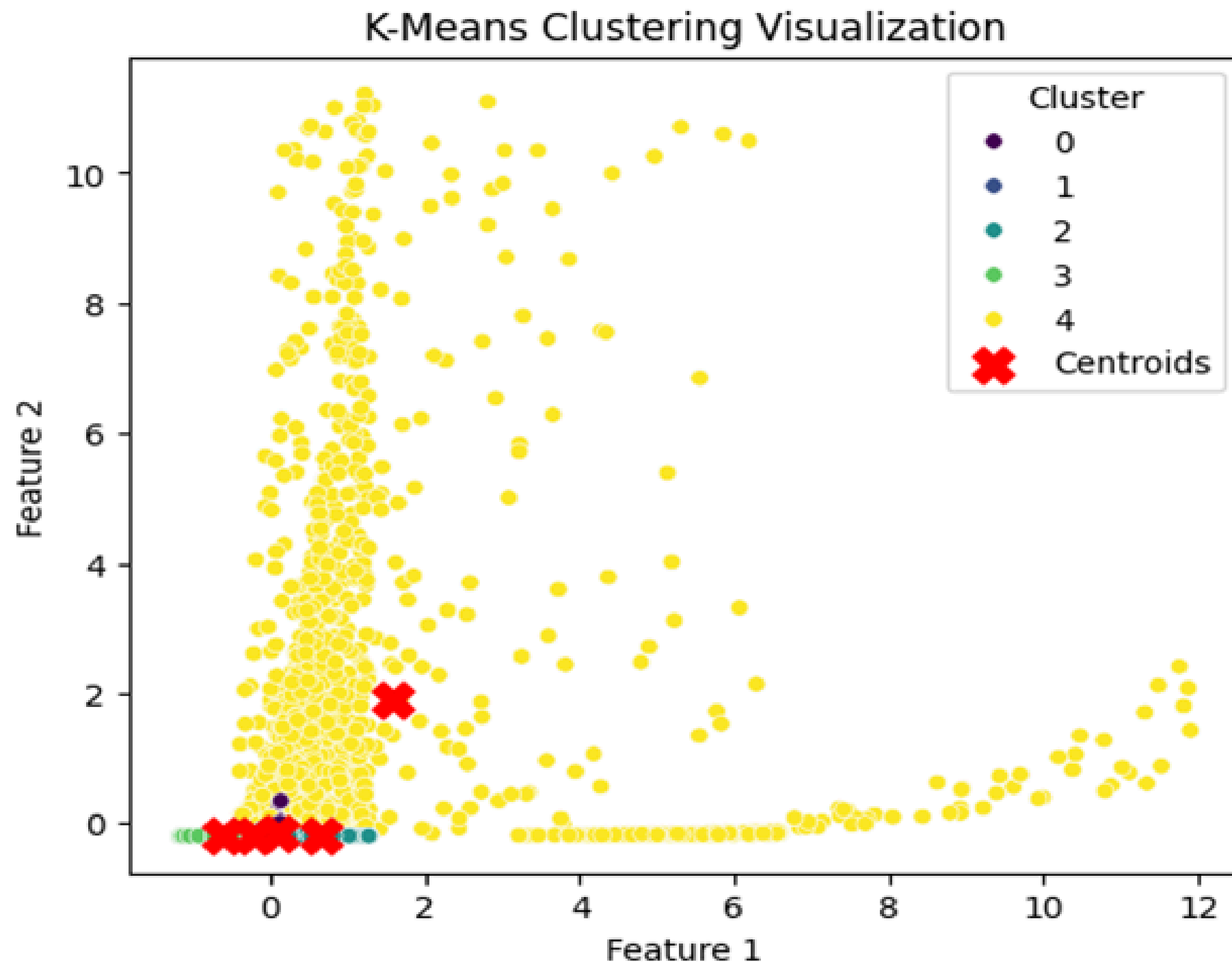


- Silhouette Method:
 - Tested K = 2 to 10
 - Highest Silhouette Score = 0.189 at K = 5
 - Confirms 5 clusters provide best separation



Observation: Both methods consistently identified 5 clusters as optimal.

K-Means Clustering



- **Implementation:**

- Model trained with $K = 5$
- Clustered stars by Temperature, Luminosity, Radius

- **Performance Metrics:**

- Silhouette Score $\rightarrow 0.189$
- Calinski–Harabasz Index $\rightarrow 1879.61$
- Davies–Bouldin Index $\rightarrow 1.857$

- **Insights:**

- Clusters are moderately compact and well-separated
- Identified 5 natural star groups in the dataset

Hierarchical Clustering

- **Method:**

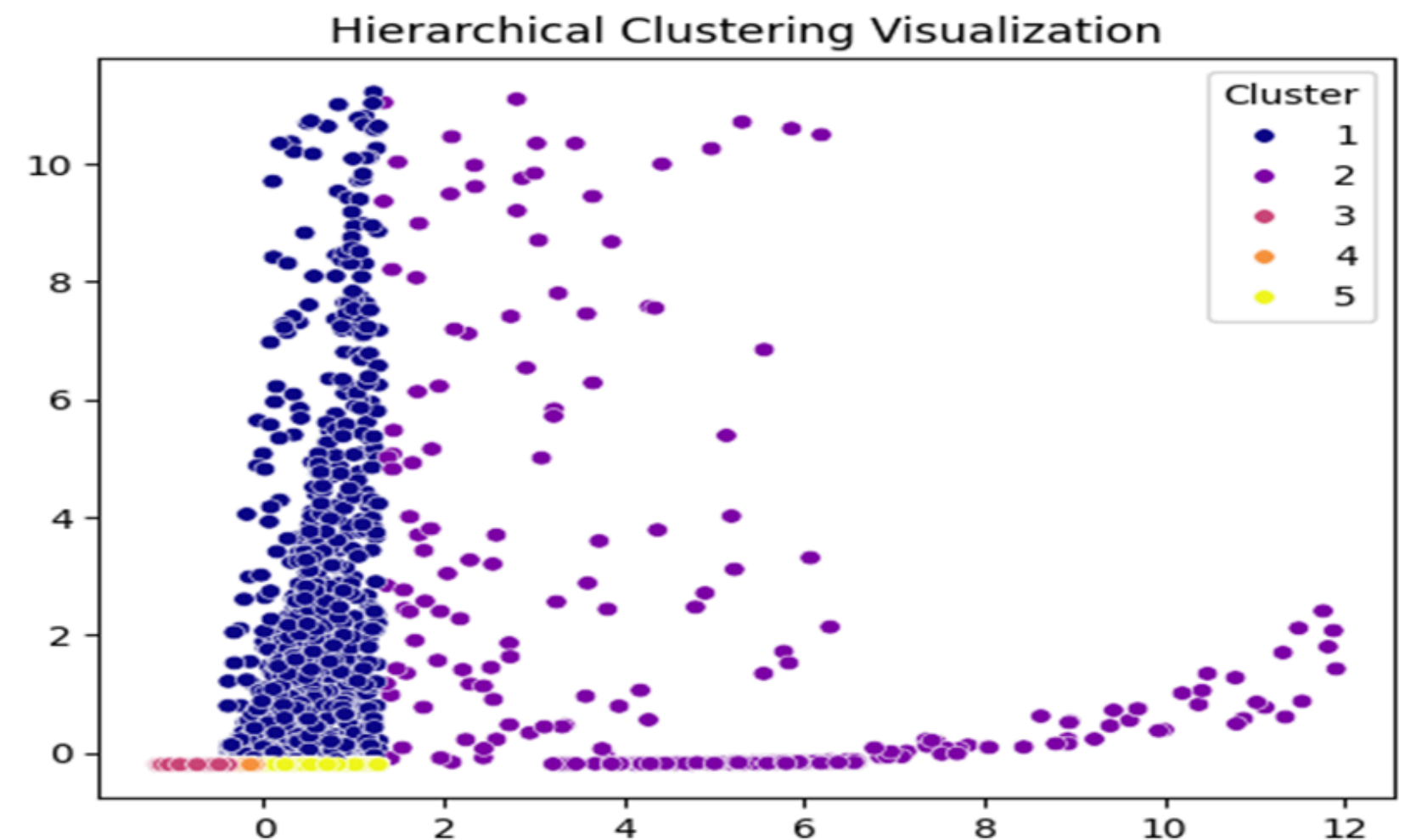
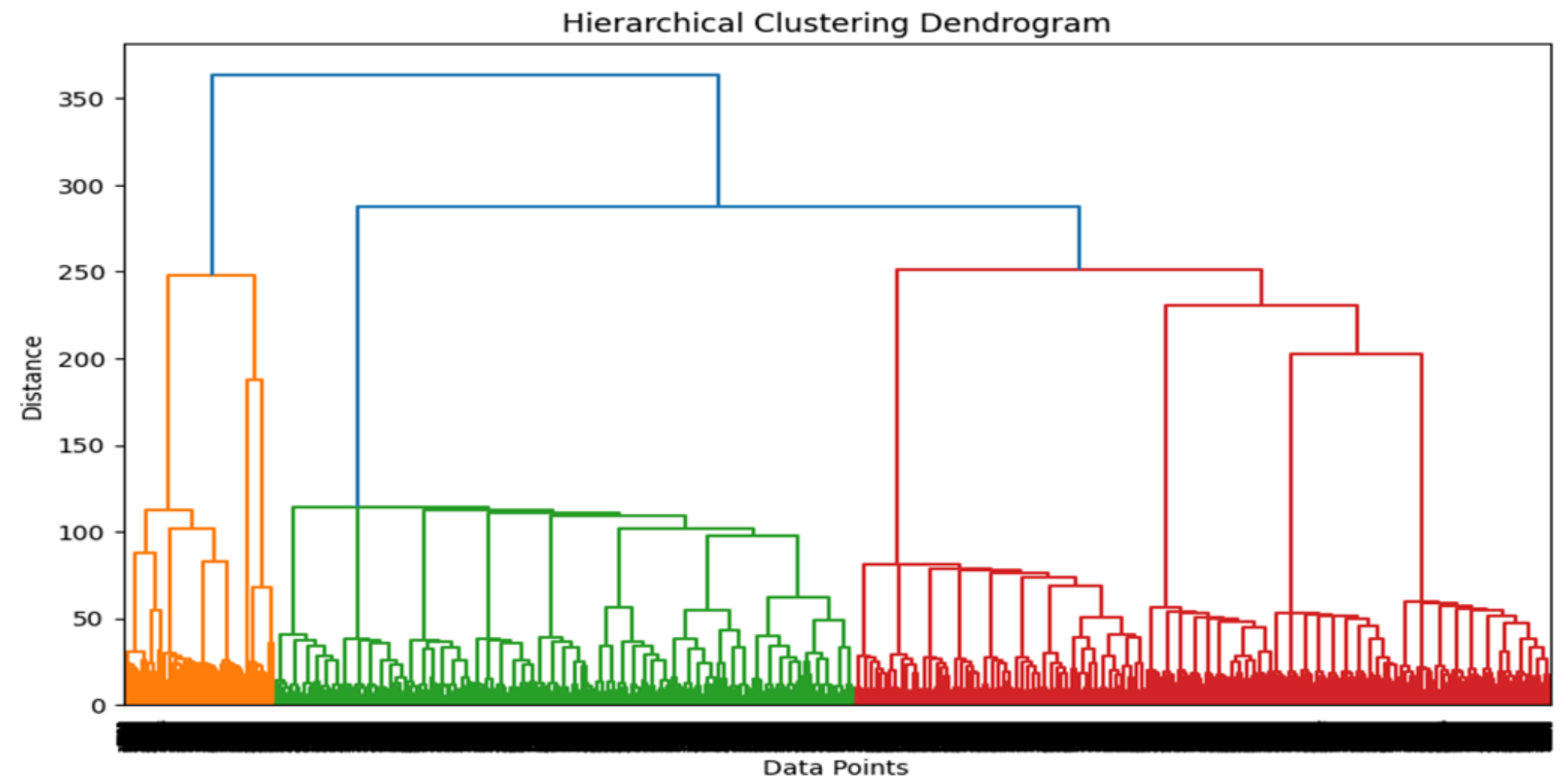
- Used Agglomerative Hierarchical Clustering with Ward Linkage
- Dendrogram revealed 5 clusters

- **Performance Metrics:**

- Silhouette Score $\rightarrow 0.171$
- Calinski–Harabasz Index $\rightarrow 1918.71$
- Davies–Bouldin Index $\rightarrow 1.925$

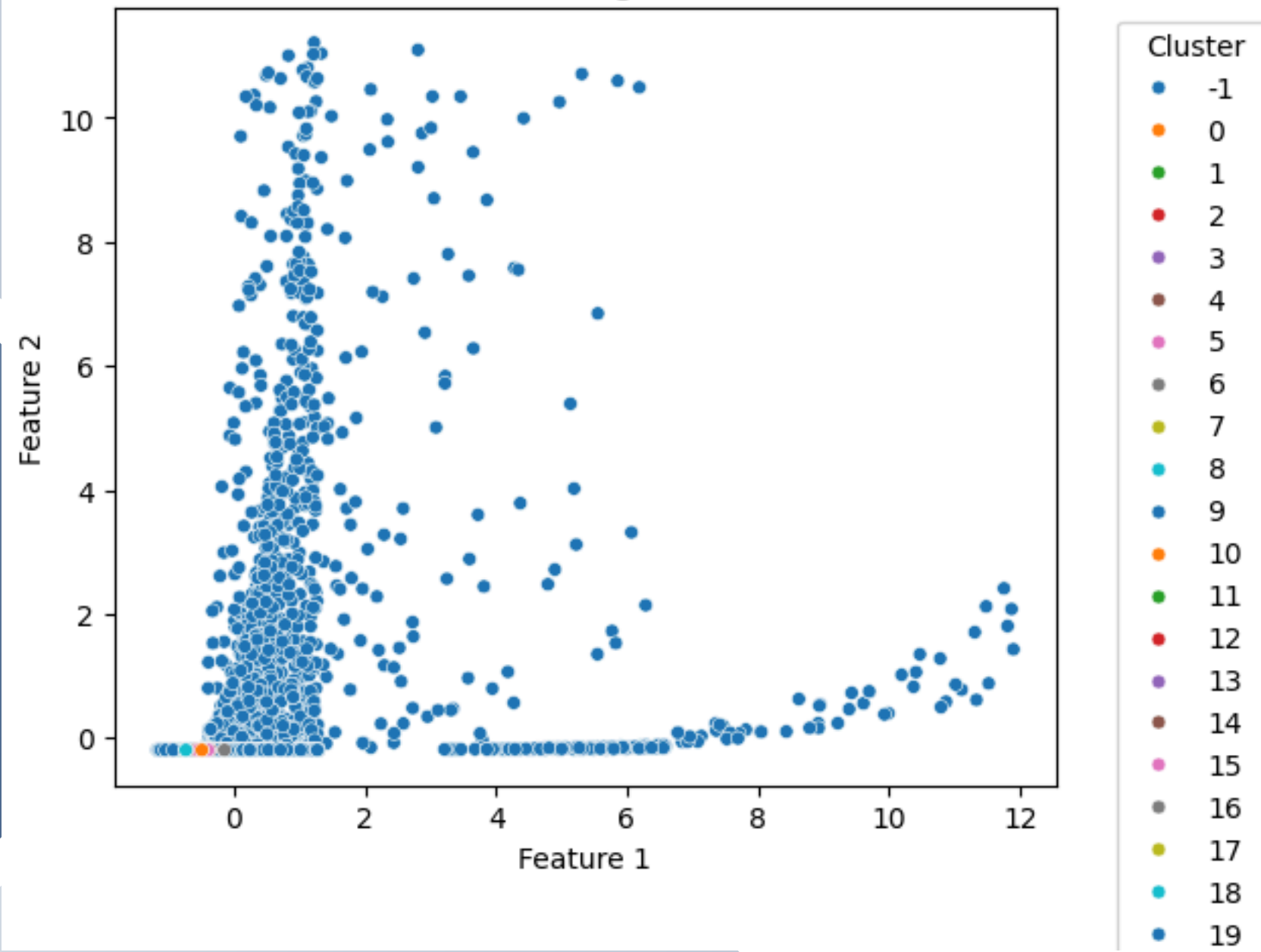
- **Observations:**

- Slightly less distinct clusters than K-Means
- Useful for visualizing relationships and nested groupings



DBSCAN Clustering

DBSCAN Clustering Visualization



- **Method:**

- Implemented Density-Based Spatial Clustering of Applications with Noise (DBSCAN).
- Automatically determined number of clusters based on data density.

- **Parameters:**

- $\text{eps} = 0.7$, $\text{min_samples} = 6$ (tuned experimentally).

- **Performance Metrics:**

- Silhouette Score $\rightarrow 0.447$
- Calinski–Harabasz Index $\rightarrow 425.94$
- Davies–Bouldin Index $\rightarrow 0.758$

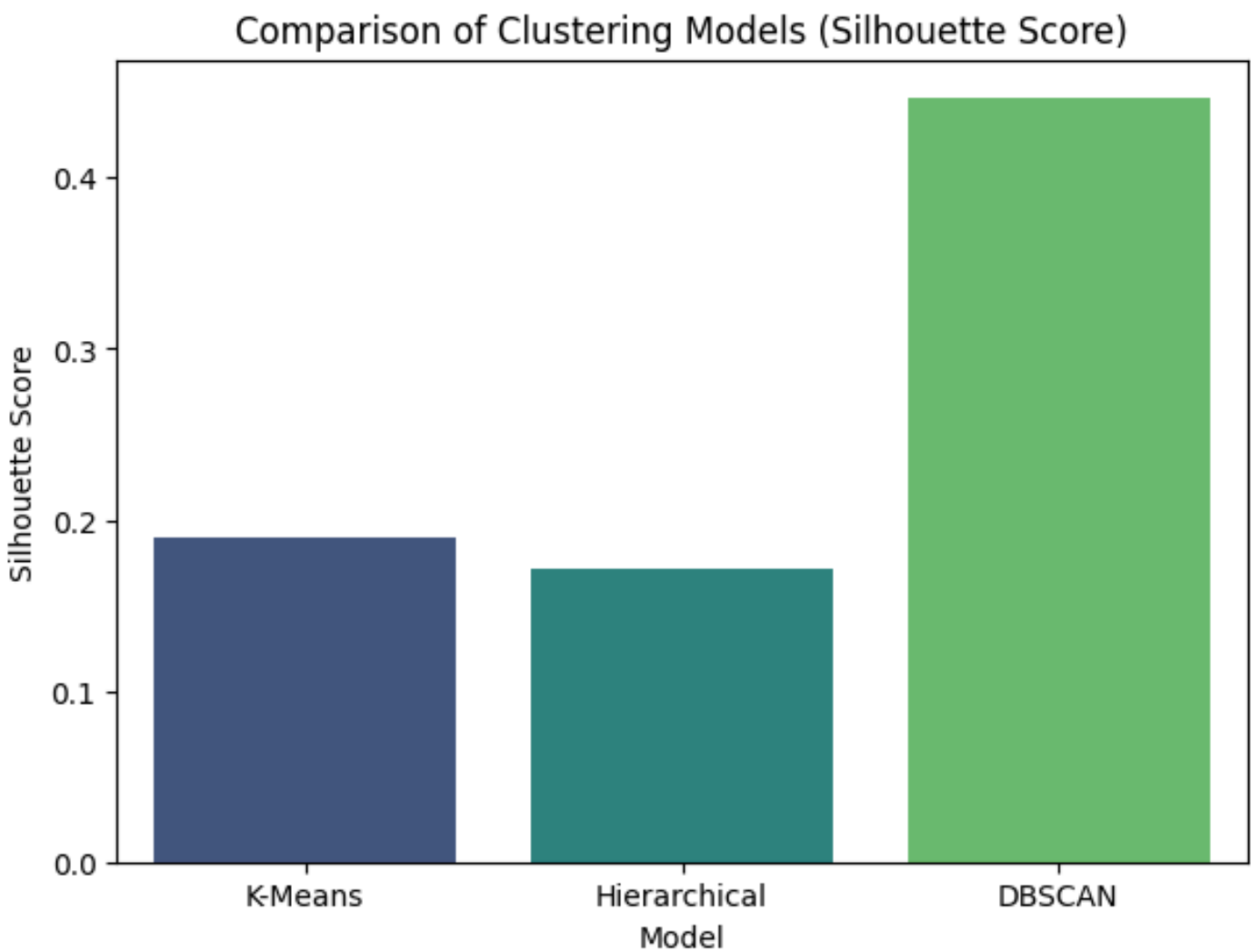
- **Insights:**

- Detected highly compact clusters with minimal overlap.
- Identified noise points (label = -1) representing outliers.
- Achieved best cluster quality among all models based on internal validation metrics.

Model Comparison & Conclusion

- **Observations:**
 - DBSCAN achieved the highest Silhouette Score, showing strong cluster compactness.
 - K-Means maintained balanced structure and interpretability.
 - Hierarchical provided clear dendrogram visualization but lower cluster distinction.
 - DBSCAN also detected outliers and density-based groups missed by other models.
- **Conclusion:**
 - DBSCAN performs best, forming distinct and noise-resistant clusters.
 - K-Means remains useful for simple, structured segmentation.
 - Overall, both show consistent results in identifying stellar groups.

Model	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
K-Means	0.189	1879.61	1.857
Hierarchical	0.171	1918.71	1.925
DBSCAN	0.447	425.94	0.758



Result Analysis

Regression Models

- Predicted Absolute Magnitude (Mv) using multiple regression techniques.
- Linear Regression served as the baseline but missed non-linear patterns.
- Polynomial Regression improved fit for curved relationships.
- Ridge, Lasso, ElasticNet gave similar results → no overfitting detected.
- Decision Tree Regression achieved near-perfect $R^2 \approx 1.0$ (possible overfitting).
- Random Forest, Gradient Boosting, and AdaBoost performed best overall.
- Random Forest Regression was the most accurate and stable ($R^2 = 1.000$).

Classification Models

- Algorithms used: Logistic Regression, Decision Tree, Random Forest, SVM, KNN.
- Preprocessing: One-hot encoding and feature scaling improved accuracy.
- Evaluated with Accuracy, Precision, Recall, and F1-Score.
- Random Forest Classifier achieved highest accuracy ($\approx 90.9\%$), followed by SVM.
- Ensemble and kernel-based models outperformed linear/distance-based ones.
- Classification confirmed preprocessing and scaling were effective.

Result Analysis

Clustering Models

- Unsupervised algorithms: K-Means, Hierarchical, and DBSCAN.
- Optimal clusters determined using Elbow Method and Silhouette Score → $k = 5$.
- K-Means: Silhouette = 0.189 → moderately defined clusters.
- Hierarchical: Silhouette = 0.171 → weaker separation.
- DBSCAN: Silhouette = 0.446 → best-defined clusters with noise handling.
- Formed meaningful star groups based on temperature and luminosity, supporting supervised model findings.

Summary of Best Performing Models

Model Type	Best Algorithm	R ² / Accuracy	Key Observation
Regression	Random Forest	1	Excellent non-linear fit
Classification	Random Forest	0.909	Most accurate classifier
Clustering	DBSCAN	Silhouette = 0.446	Best-defined clusters

Conclusion

- Machine learning was applied to analyze and predict star types using temperature, luminosity, radius, and magnitude.
- Preprocessing and feature scaling improved data quality and model accuracy.
- Supervised (Regression, Classification) and Unsupervised (Clustering) models were used effectively.
- Random Forest, SVM, AdaBoost, and Gradient Boosting gave the best results.
- Ensemble models showed higher accuracy and less overfitting.
- The project proved that data-driven methods can uncover patterns and enhance understanding of stellar behavior.



Thank You